

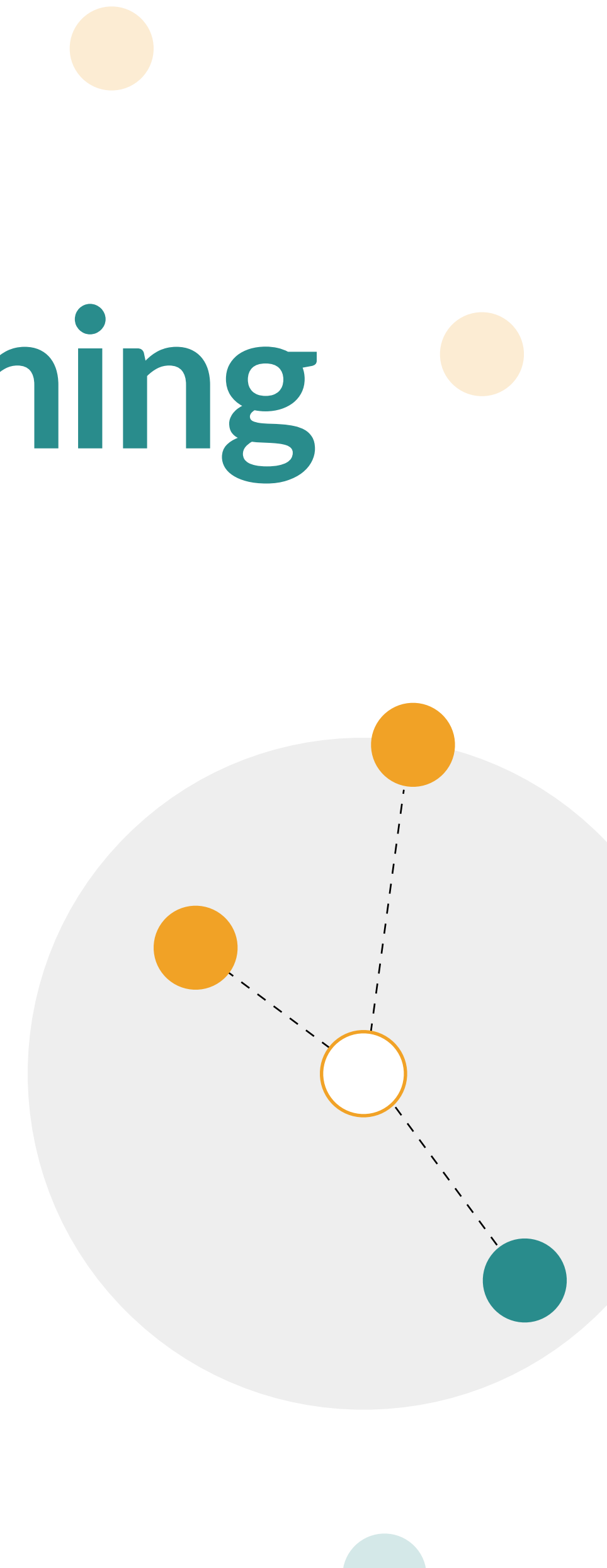
Foundations of Machine Learning

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Lecture 3 : Regression

Master's Degree in **AI & Society** (PSL-PSAI, Fall 2026-27)

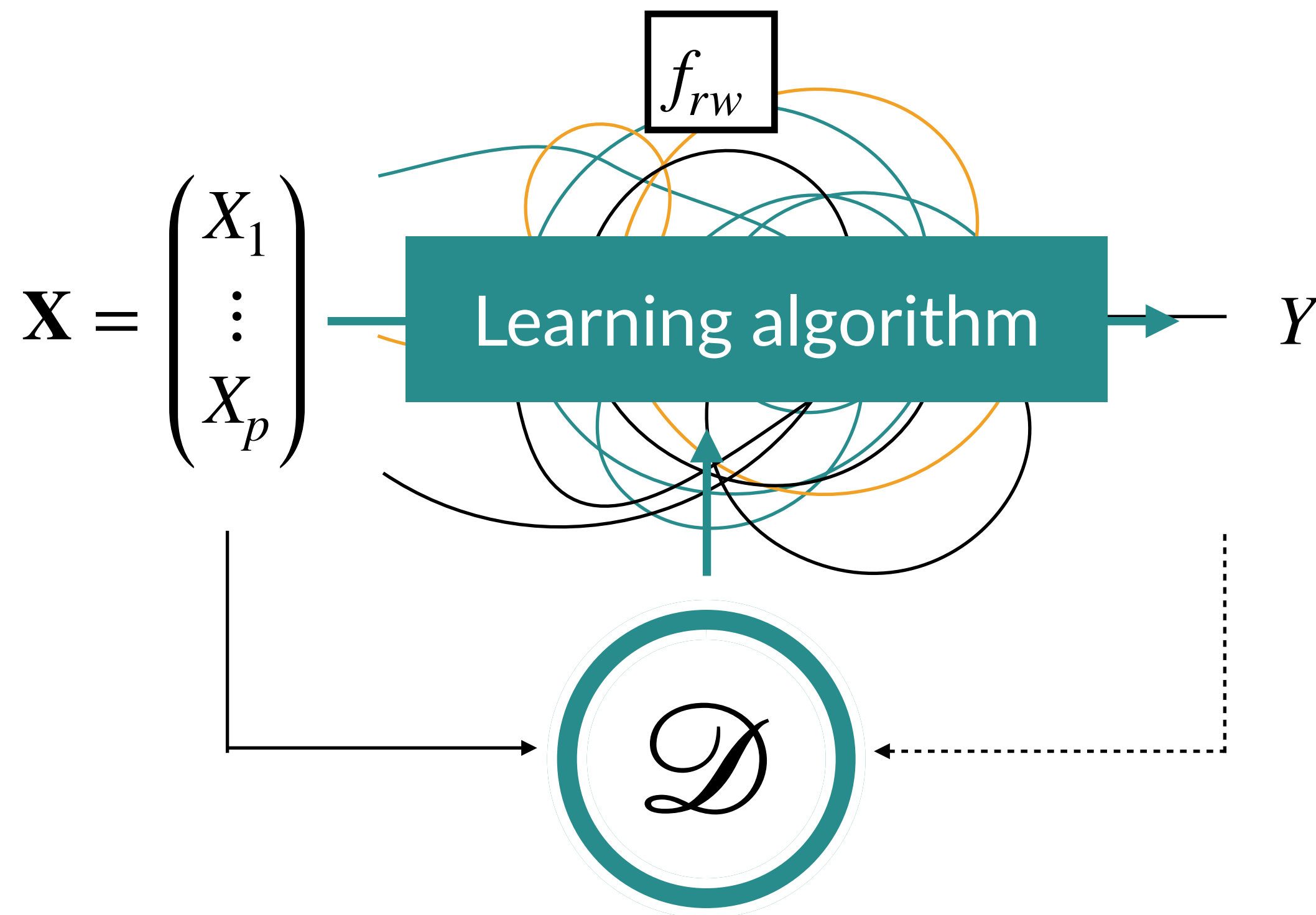


Last week

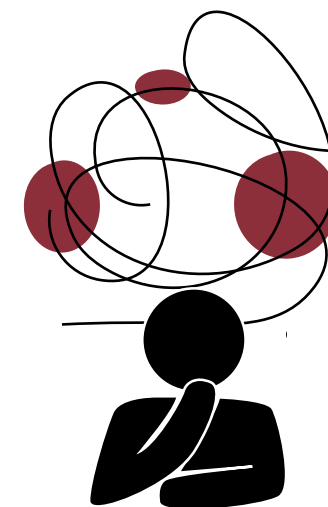
Statistical Learning Framework



We have a set of variables $\mathbf{X} \in \mathcal{X}$ —the predictors, or features—and a quantity $Y \in \mathcal{Y}$ we want to predict. They are related by some unknown function $f_{rw} : \mathcal{X} \rightarrow \mathcal{Y}$. Our goal is to approximate f_{rw} .

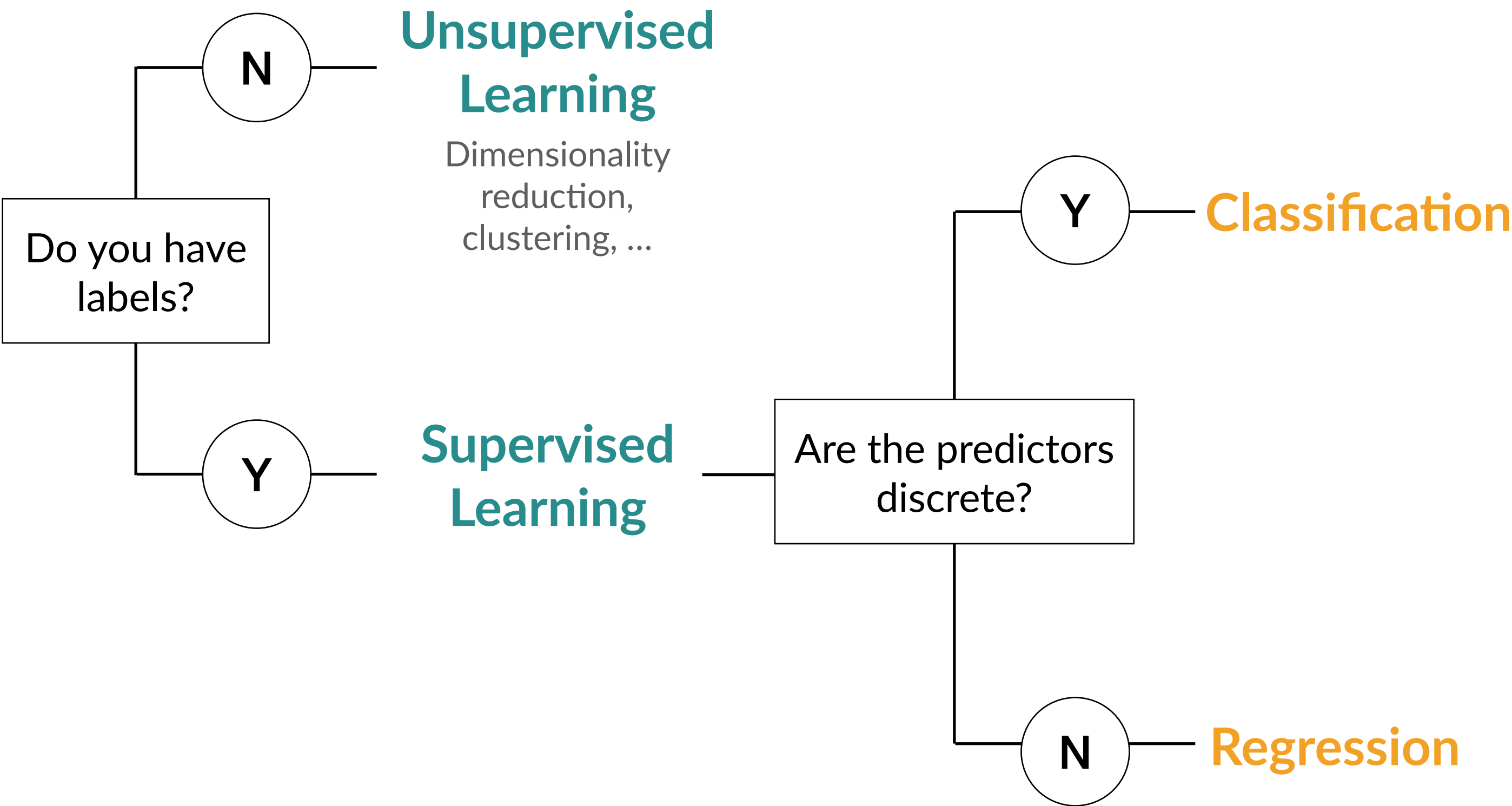


Collect data $\mathcal{D}$ and use a learning algorithm to build an approximation of f_{rw} .

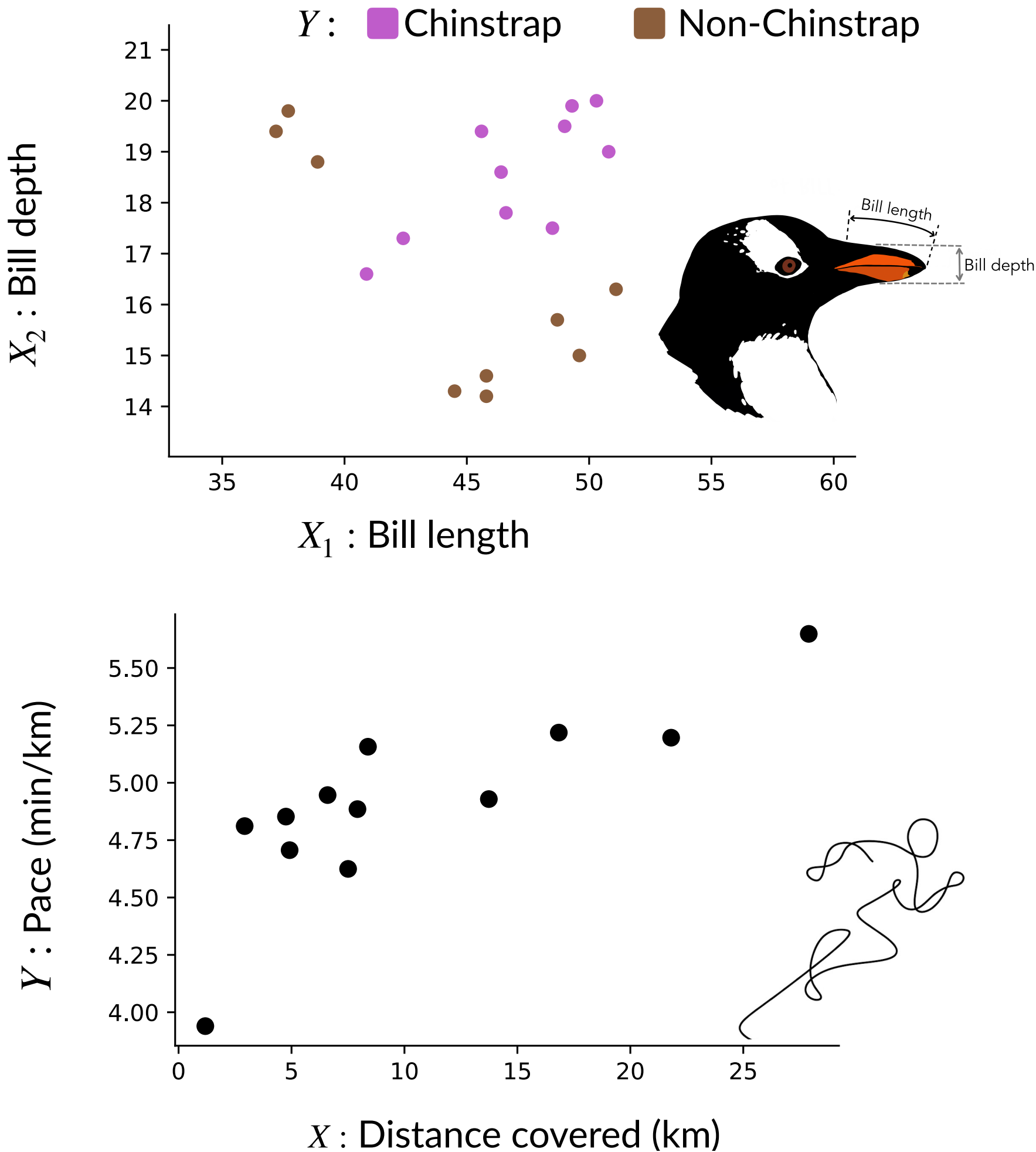


However, we only ever observe **a finite, noisy sample**

Learning Paradigms



Other dichotomies /
Goal : **prediction** vs **inference**
Method : **parametric** vs **non-parametric**



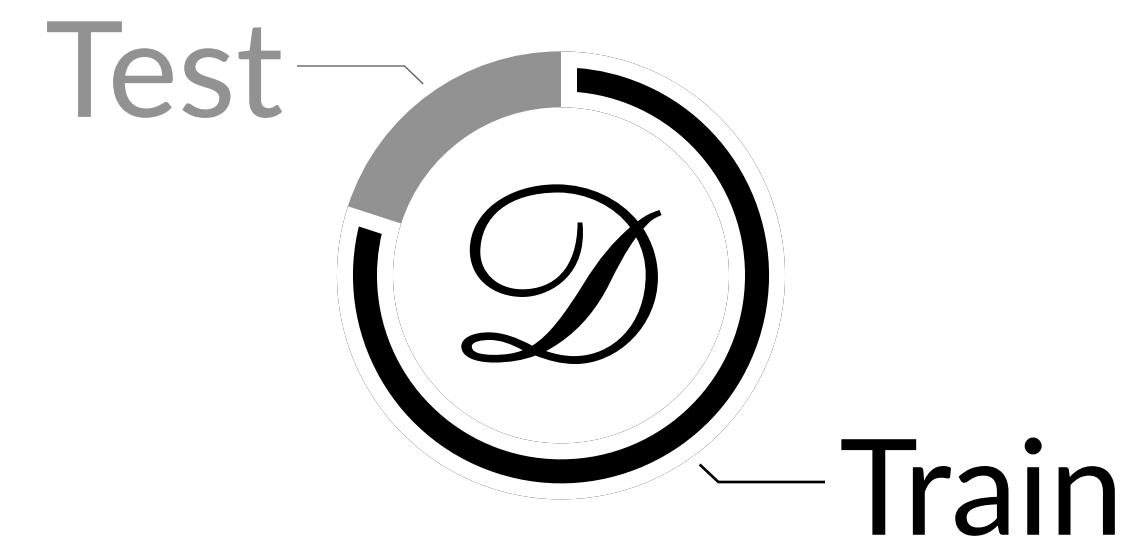
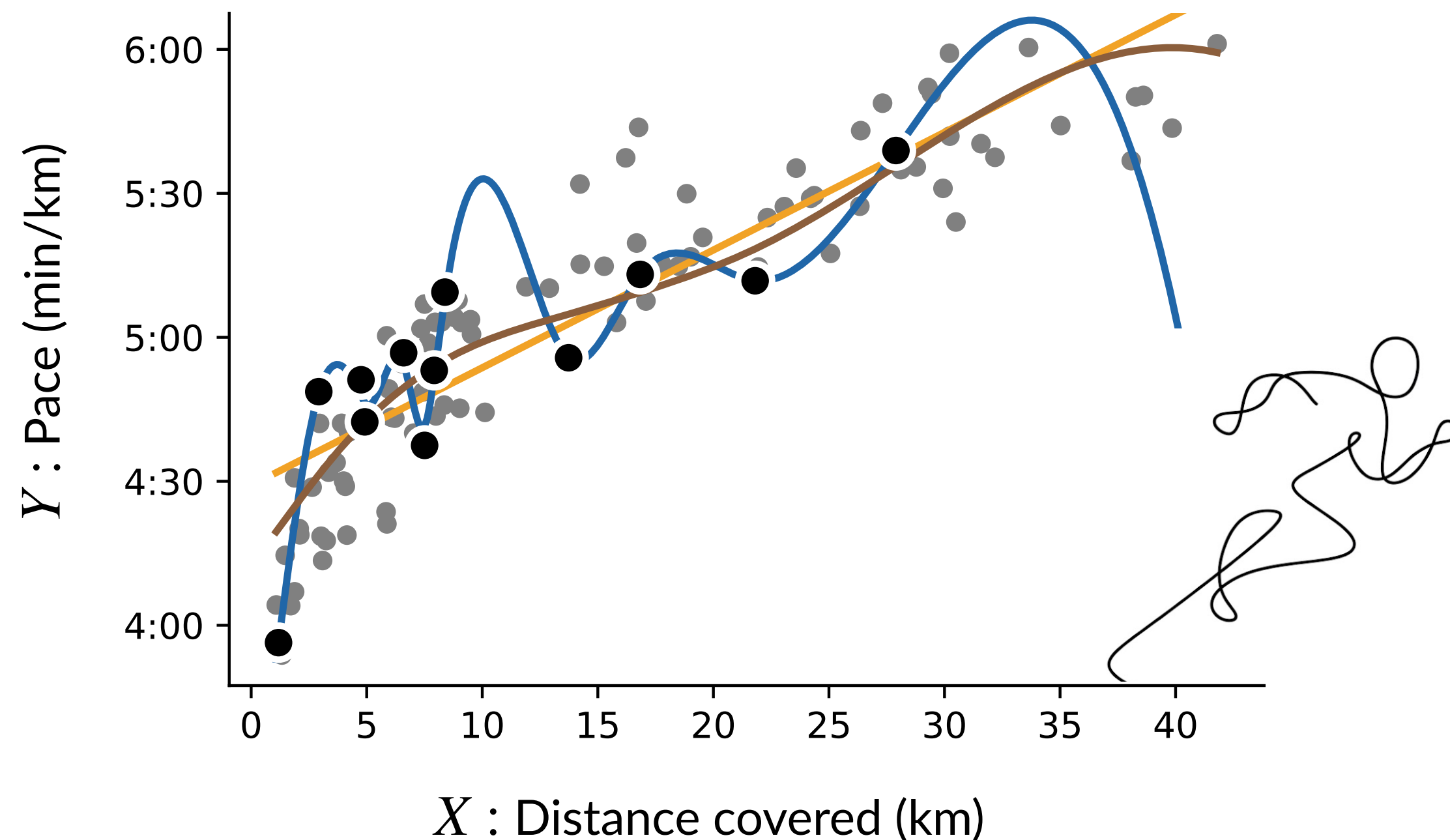
Error & Generalisation

We need a precise, quantitative way to measure how good the predictions are. This is the role of the **error**, or **loss**, function.

We want $\hat{f}$ that performs well on new data. We seek the smallest possible **test error**

$$\text{MSE}_{\text{test}} = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} (y_i - \hat{f}(x_i))^2$$

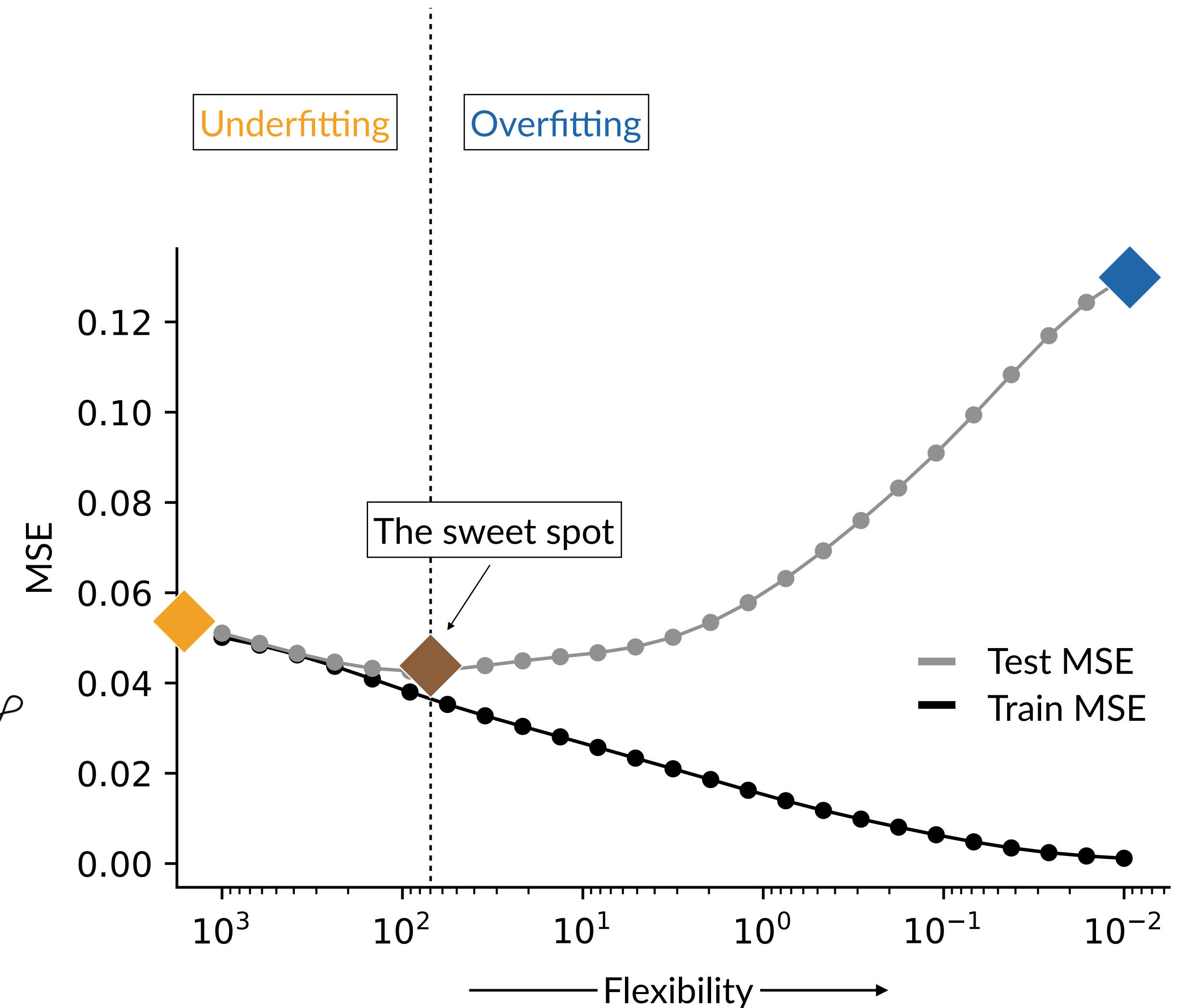
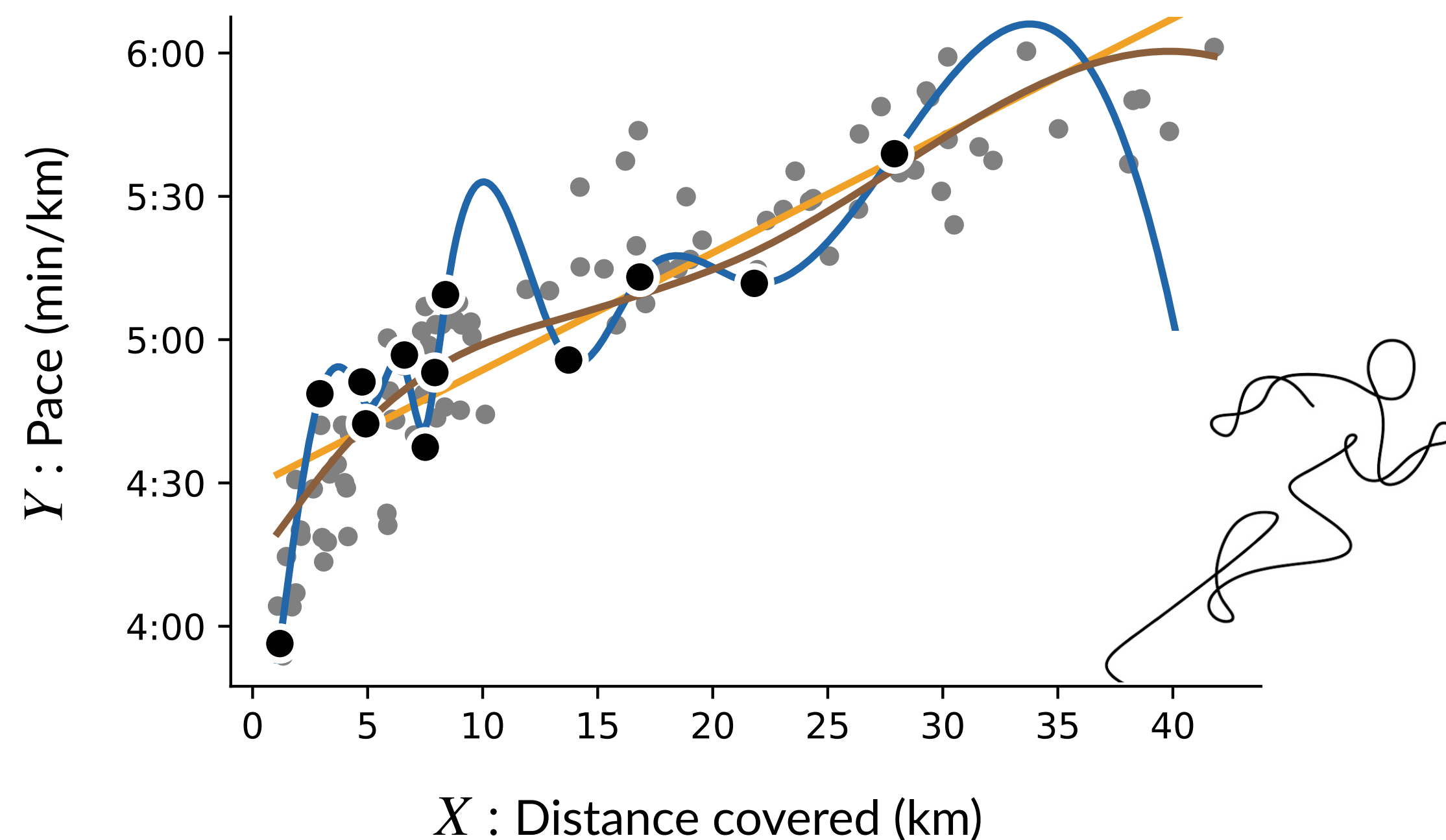
where $(x_i, y_i) \in \mathcal{D}_{\text{test}}$ are observations not used to train, or fit, $\hat{f}$



Naming the Tension

U-shaped Test MSE

How does **test error** behave as model **flexibility** increases?



Decision Theory

The Prediction Problem

Setup

Let $\mathbf{X} \in \mathbb{R}^p$ be a random input vector and $Y \in \mathbb{R}$ an outcome

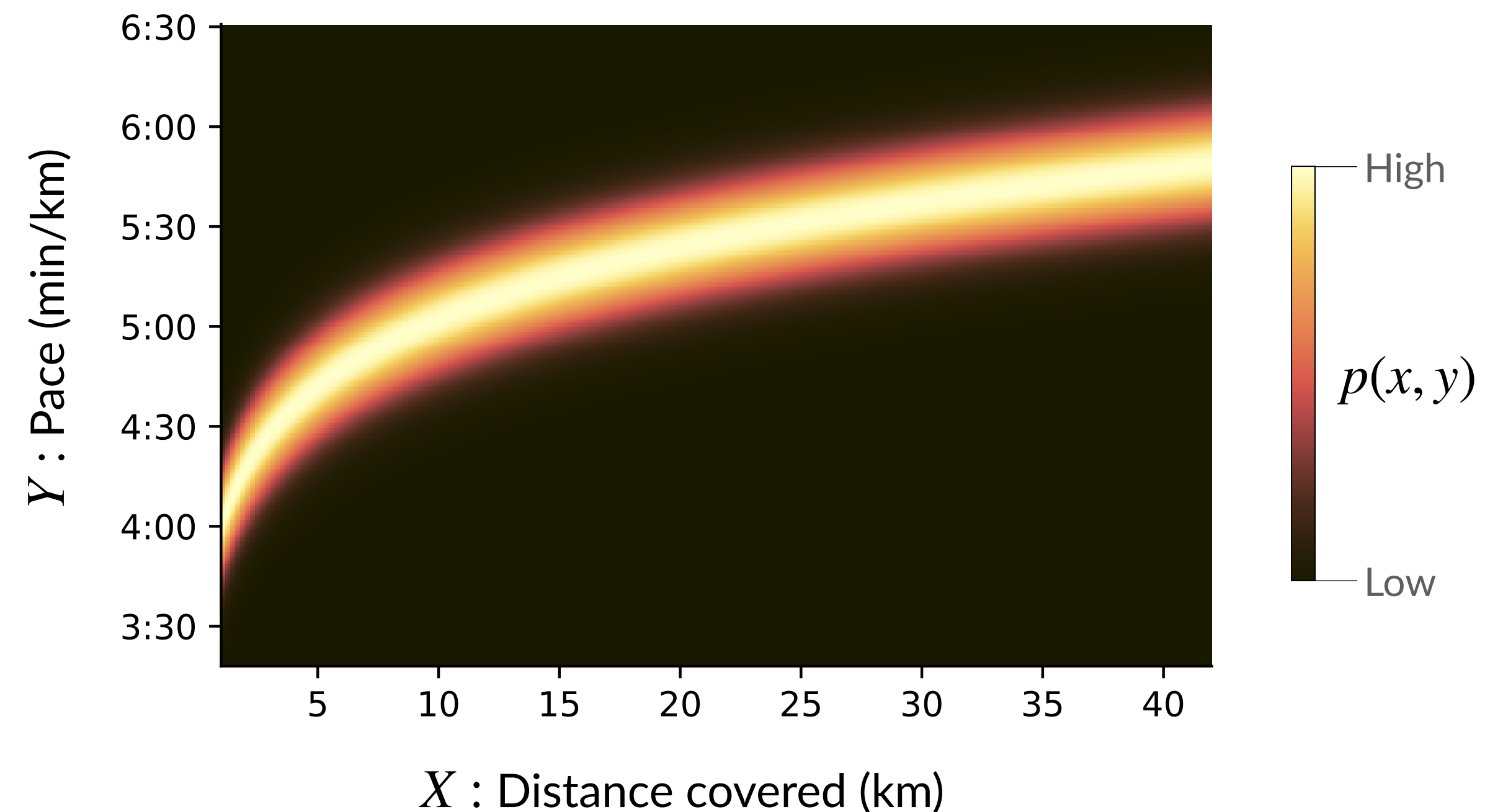
Assumption

The joint density $p(x, y)$ is known exactly

Unrealistic in practice, but a useful thought experiment: its purpose is to define the best prediction, independently of any data

This section
is data-free

Population joint density of distance and pace (brighter=more probable).
At each distance, pace is spread over a range of values



The Prediction Problem

Setup

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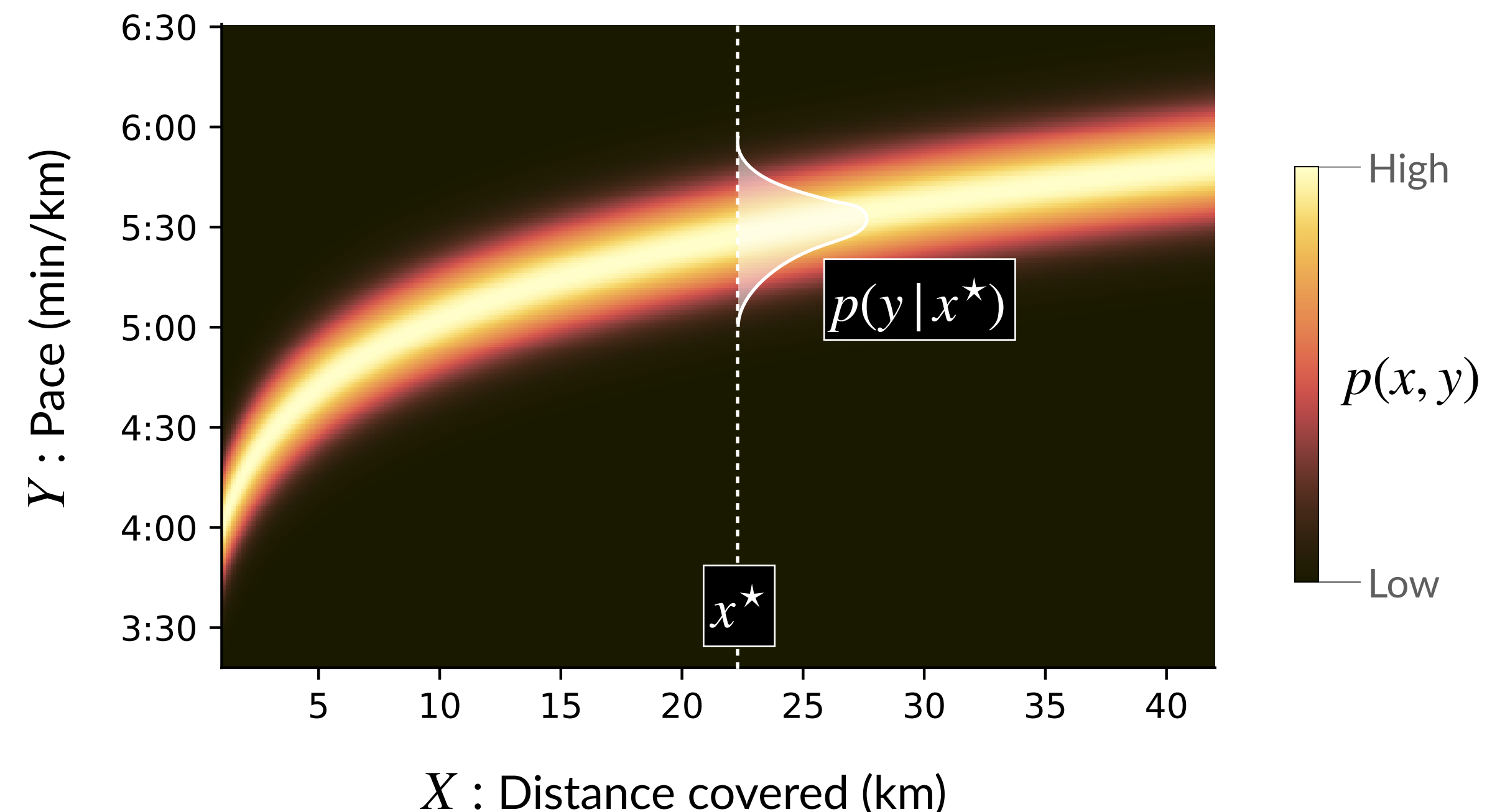
Task

We seek a function $f(X)$ for predicting Y given values of $\mathbf{X}$

Any function, no restriction yet

Population joint density of distance and pace (brighter=more probable).
At each distance, pace is spread over a range of values.

For a value $x^\star$, many y are possible, spread according to $p(y | x^\star)$.
Which **decision** on the single value to use a « prediction » ?
Mean of $p(y | x^\star)$? Median? Max? Other?



The Loss Function

Measuring the Cost of a Wrong Prediction

A **loss function** is a function $\mathcal{L} : \mathbb{R} \times \mathbb{R} \rightarrow [0, \infty)$ that gives the cost $\mathcal{L}(y, \hat{y})$ of predicting $\hat{y}$ when the true value is y . It is chosen by us and determines what « best » means

The same distribution with a different loss
gives a different best prediction

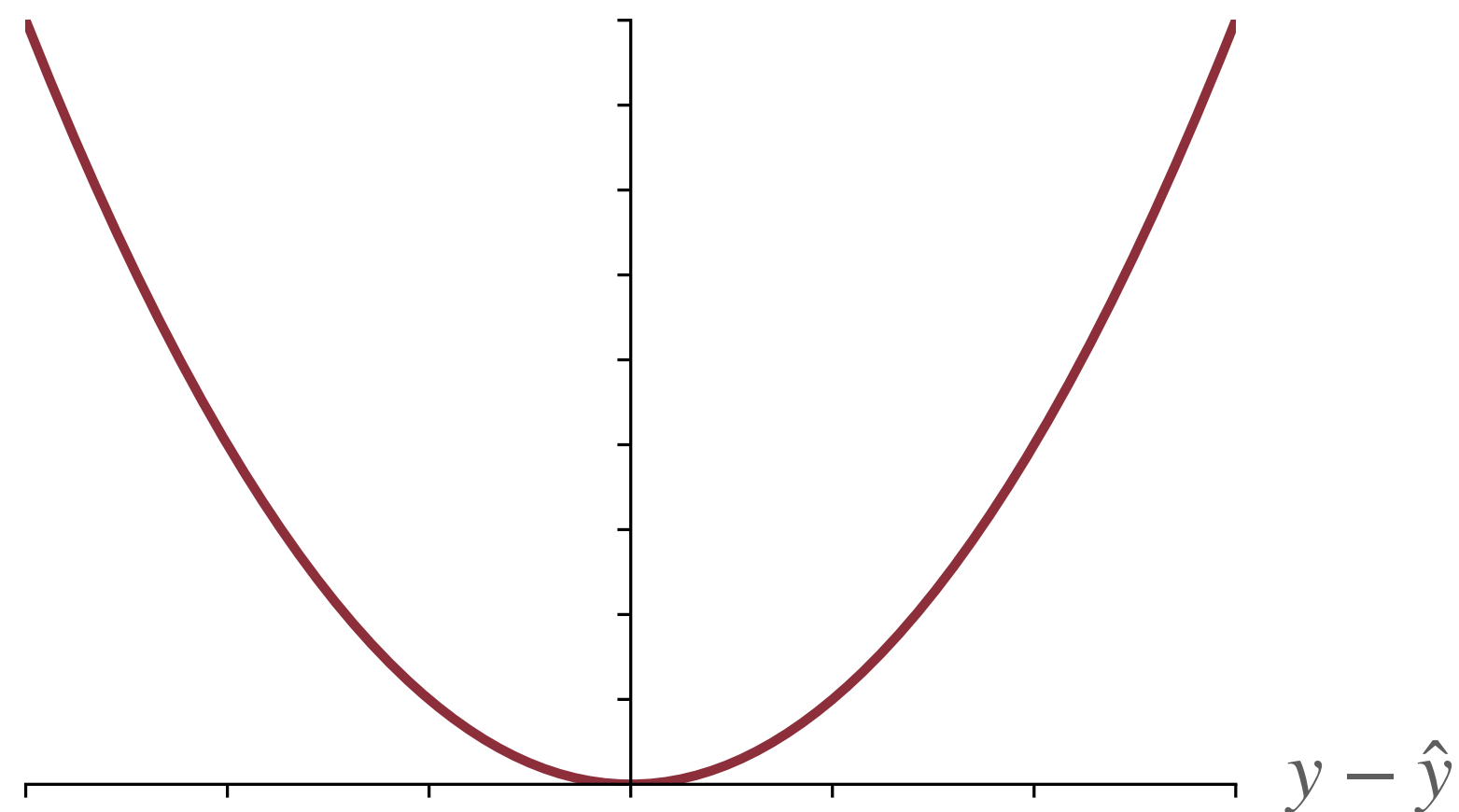
The Loss Function

Measuring the Cost of a Wrong Prediction

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Squared Loss

$$\mathcal{L}(y, \hat{y}) = (y - \hat{y})^2$$



Behaviour

Zero for a perfect prediction and depends only on the size of the miss, not on its direction

Sensitivity

Large errors dominate. Twice the miss costs four times as much

Symmetry

Overshooting and undershooting by the same amount cost the same

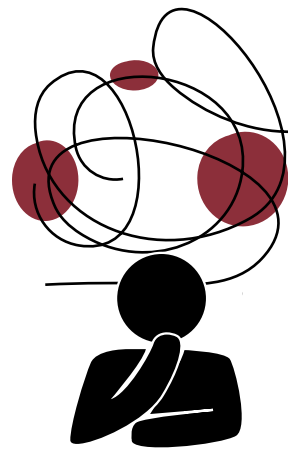
Convenience

Smooth with a single minimum, which makes the math easy

Other choices are possible, for example $\mathcal{L}(y, \hat{y}) = |y - \hat{y}|$

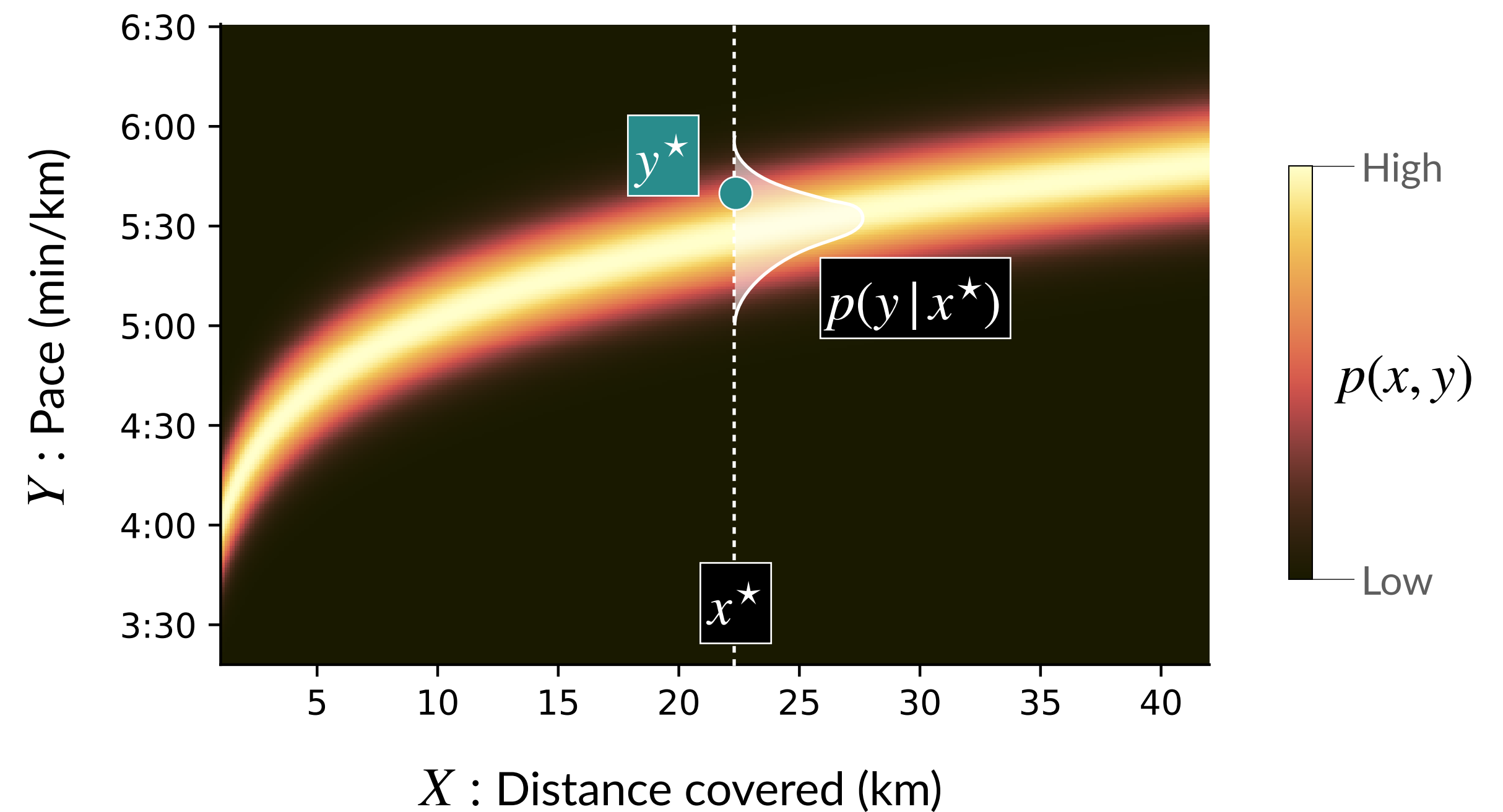
The Risk

Averaging the Loss



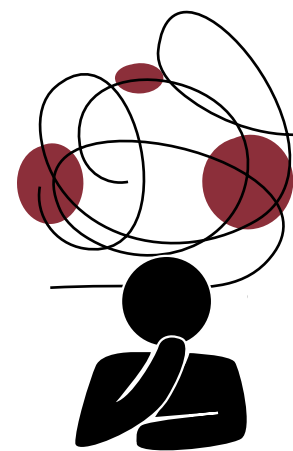
Consider a candidate $f(\mathbf{X})$. The loss $\mathcal{L}(Y, f(\mathbf{X}))$ is random, because it is function of random variables $(\mathbf{X}, Y)$

Two candidates can't be compared by one random outcome, and one lucky draw proves nothing



The Risk

Averaging the Loss



Consider a candidate $f(\mathbf{X})$. The loss $\mathcal{L}(Y, f(\mathbf{X}))$ is random, because it is function of random variables $(\mathbf{X}, Y)$

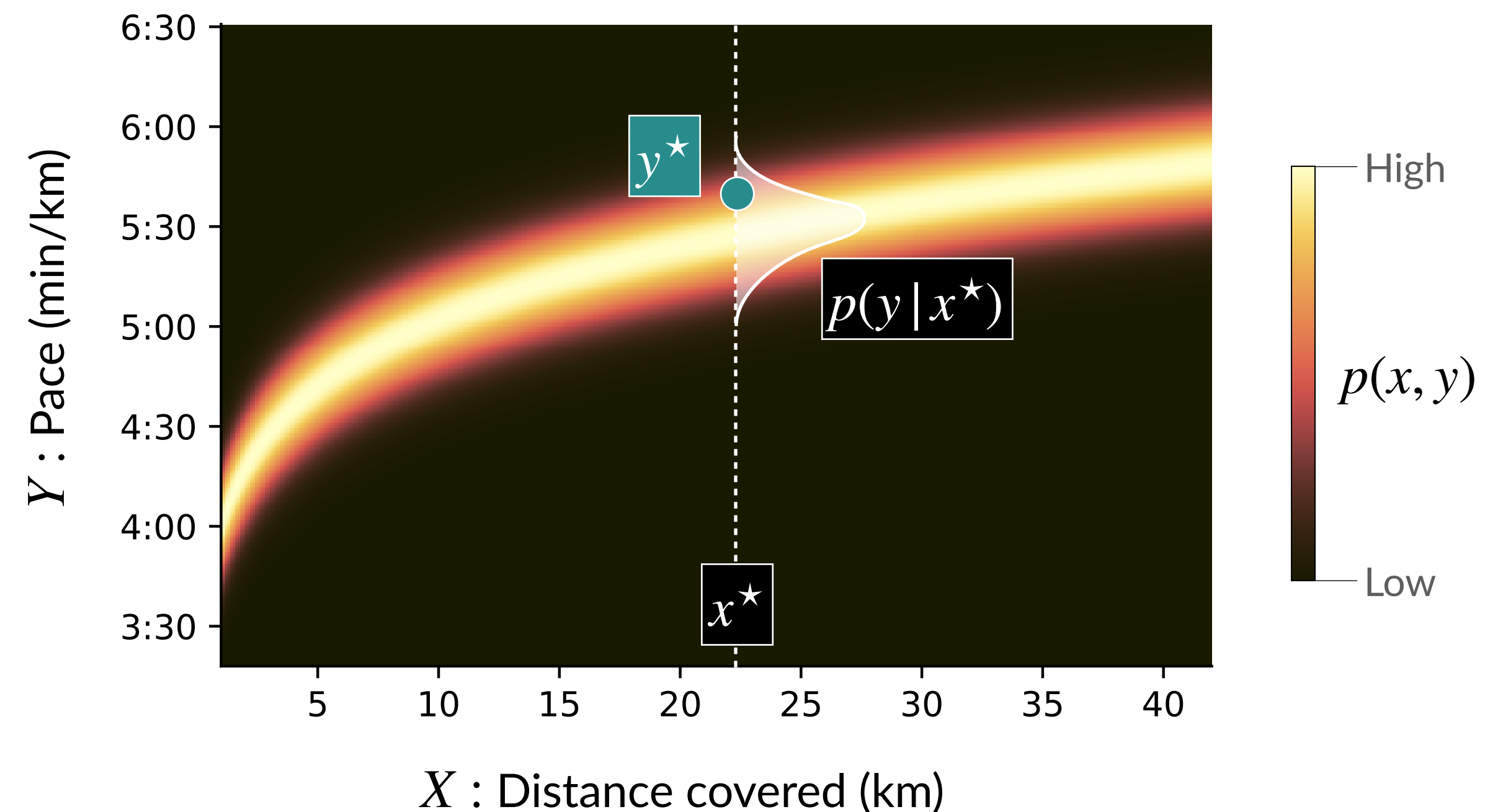
Two candidates can't be compared by one random outcome, and one lucky draw proves nothing



Summarise the random loss by its average over the population. This is the **risk**:

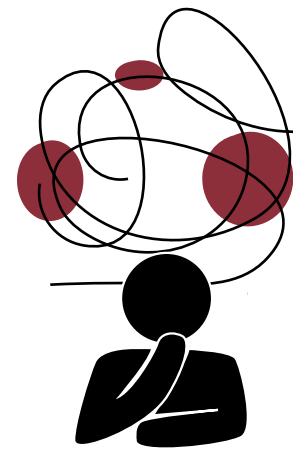
$$\begin{aligned} R(f) &\equiv \mathbb{E} \left[L(Y, f(\mathbf{X})) \right] \\ &= \int_{\mathbb{R}^p} \int_{\mathbb{R}} L(y, f(\mathbf{x})) p(\mathbf{x}, y) dy d\mathbf{x} \end{aligned}$$

The average cost we would pay if we used f on many new pairs $(\mathbf{X}, Y)$. One value per candidate, so different candidates can now be compared



The Risk

Optimisation principle



Consider a candidate $f(\mathbf{X})$. The loss $\mathcal{L}(Y, f(\mathbf{X}))$ is random, because it is function of random variables $(\mathbf{X}, Y)$

Two candidates can't be compared by one random outcome, and one lucky draw proves nothing



Summarise the random loss by its average over the population. This is the **risk**:

$$R(f) \equiv \mathbb{E} [L(Y, f(\mathbf{X}))]$$
$$= \int_{\mathbb{R}^p} \int_{\mathbb{R}} L(y, f(\mathbf{x})) p(\mathbf{x}, y) dy d\mathbf{x}$$

Risk Minimisation

$$f^* = \arg \min_f R(f)$$

└ No restriction on f

Depends on the choice of the loss and on the joint distribution

$\Rightarrow$ « Best » f = smallest risk

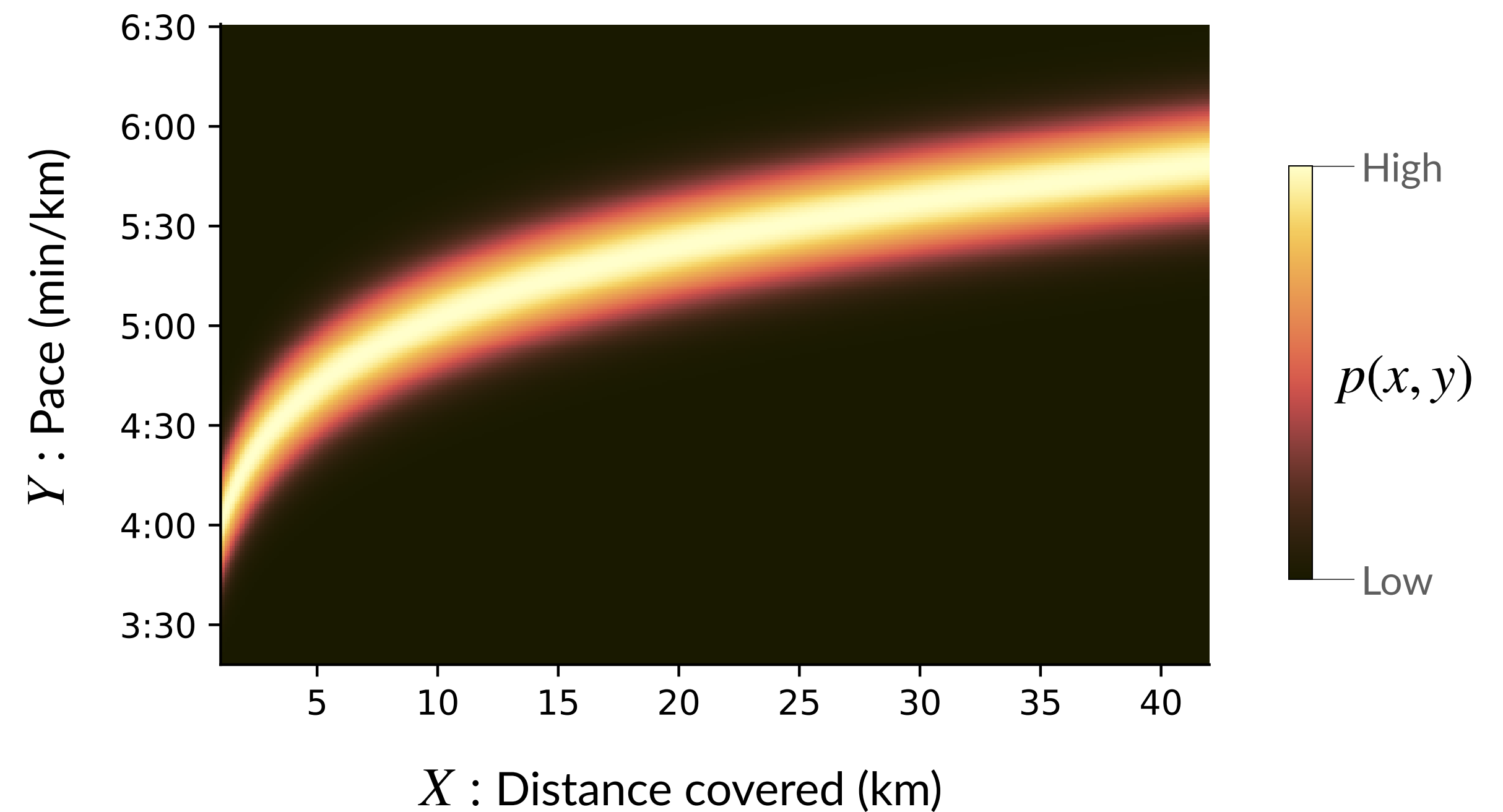
The risk goes by several names. Sometimes called the « expected loss », or the « expected prediction error » (EPE) in ESL

Risk Minimisation with Squared Loss

Task

Solve the **Risk Minimisation** problem for the **Squared Loss**

Find the function f^* with the smallest expected squared miss



Risk Minimisation with Squared Loss

Derivation

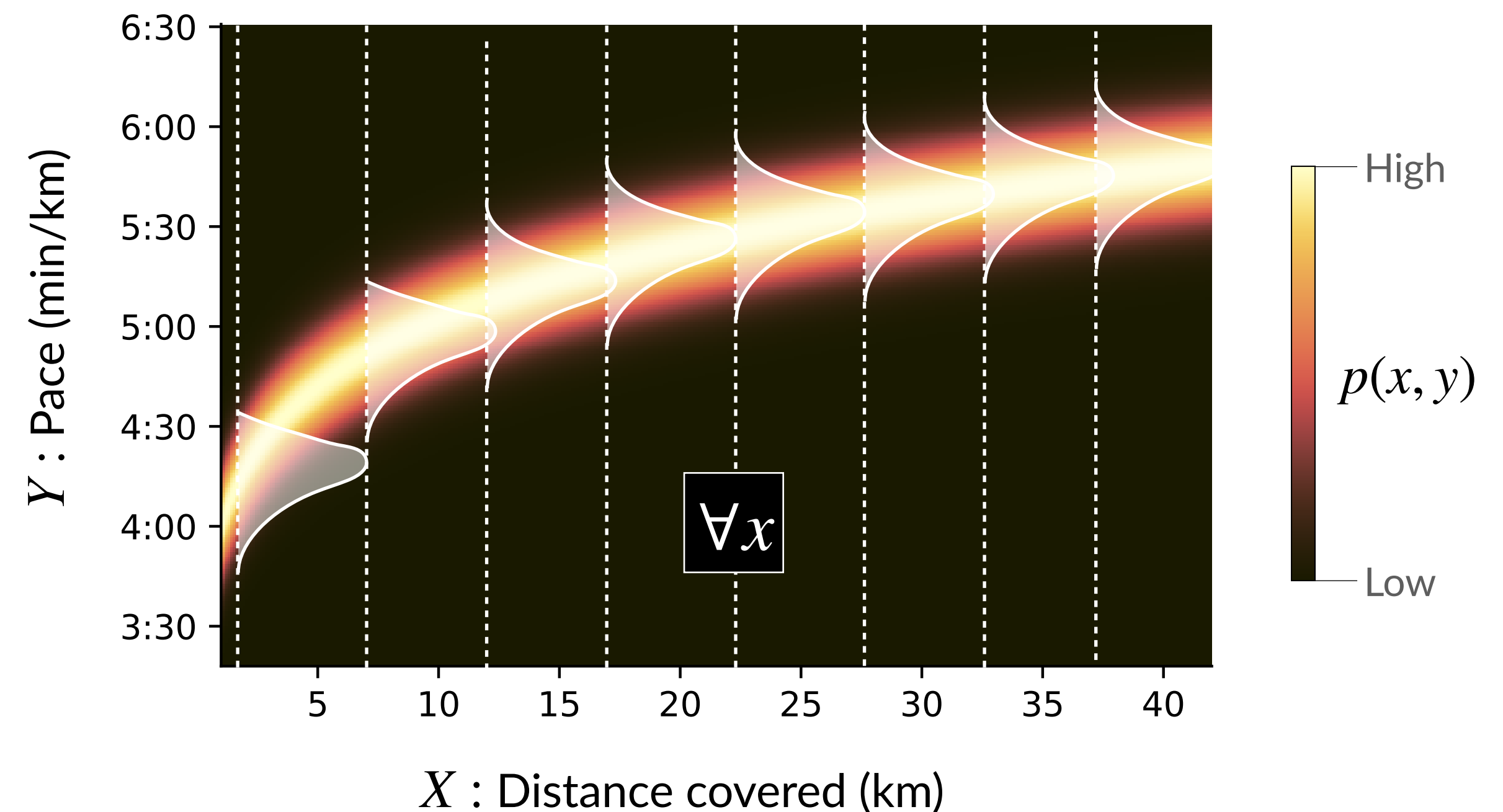
$$\begin{aligned} R(f) &= \mathbb{E} \left[(Y - f(\mathbf{X}))^2 \right] \\ &= \int_{\mathbb{R}^p} \int_{\mathbb{R}} (y - f(\mathbf{x}))^2 p(\mathbf{x}, y) dy d\mathbf{x} \quad \text{where } p(\mathbf{x}, y) = p(y | \mathbf{x}) p(\mathbf{x}) \\ &= \int_{\mathbb{R}^p} \left[\int_{\mathbb{R}} (y - f(\mathbf{x}))^2 p(y | \mathbf{x}) dy \right] p(\mathbf{x}) d\mathbf{x} \\ &= \mathbb{E}_{\mathbf{X}} \left[\mathbb{E}_{Y|\mathbf{X}} \left[(Y - f(\mathbf{X}))^2 \mid \mathbf{X} \right] \right] \end{aligned}$$

The inner expectation is the expected cost at a fixed $\mathbf{x}$. The outer expectation just averages these costs over $\mathbf{x}$. So we can minimise the inner cost point-wise, that is, separately at each $\mathbf{x}$

Task

Solve the **Risk Minimisation** problem for the **Squared Loss**

Find the function f^* with the smallest expected squared miss



Risk Minimisation with Squared Loss

Derivation

Risk Minimisation

$$f^*(x) \equiv \arg \min_c \mathbb{E}_{Y|X} \left[(Y - c)^2 \mid \mathbf{X} = \mathbf{x} \right]$$

Average over
the slice

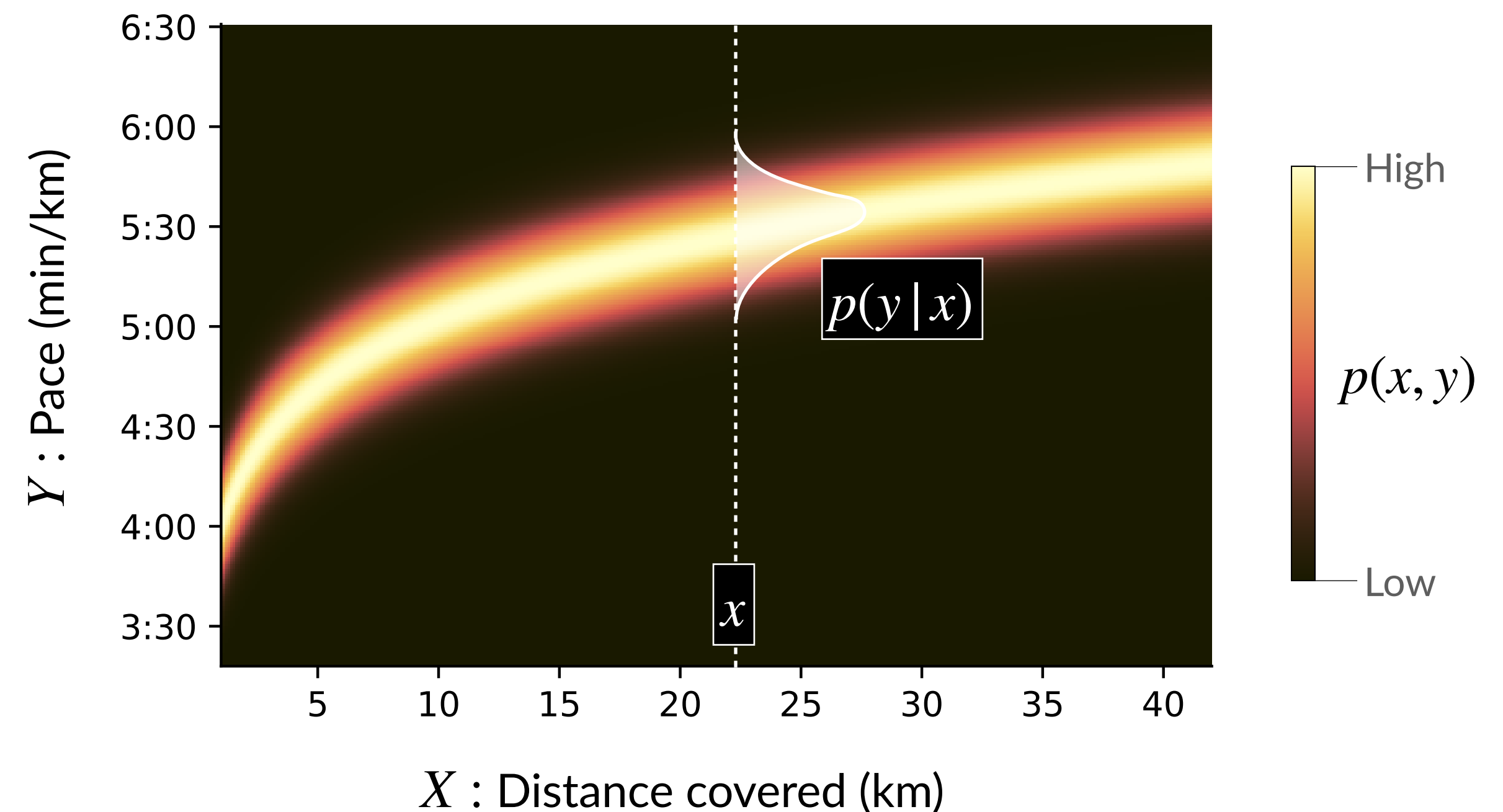
Try a value c

Pick one $\mathbf{x}$

Task

Solve the **Risk Minimisation** problem
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squared miss



Risk Minimisation with Squared Loss

Derivation

Risk Minimisation

$$f^*(x) \equiv \arg \min_c \mathbb{E}_{Y|X} [(Y - c)^2 \mid \mathbf{X} = \mathbf{x}]$$

Regression function

$$f^*(x) = \mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}]$$

Proof /

$$\begin{aligned} f^*(x) &= \arg \min_c [\mathbb{E}[Y^2 \mid X = \mathbf{x}] - 2c \mathbb{E}[Y \mid X = \mathbf{x}] + c^2] \\ &= \arg \min_c [-2c \mathbb{E}[Y \mid X = \mathbf{x}] + c^2] \end{aligned}$$

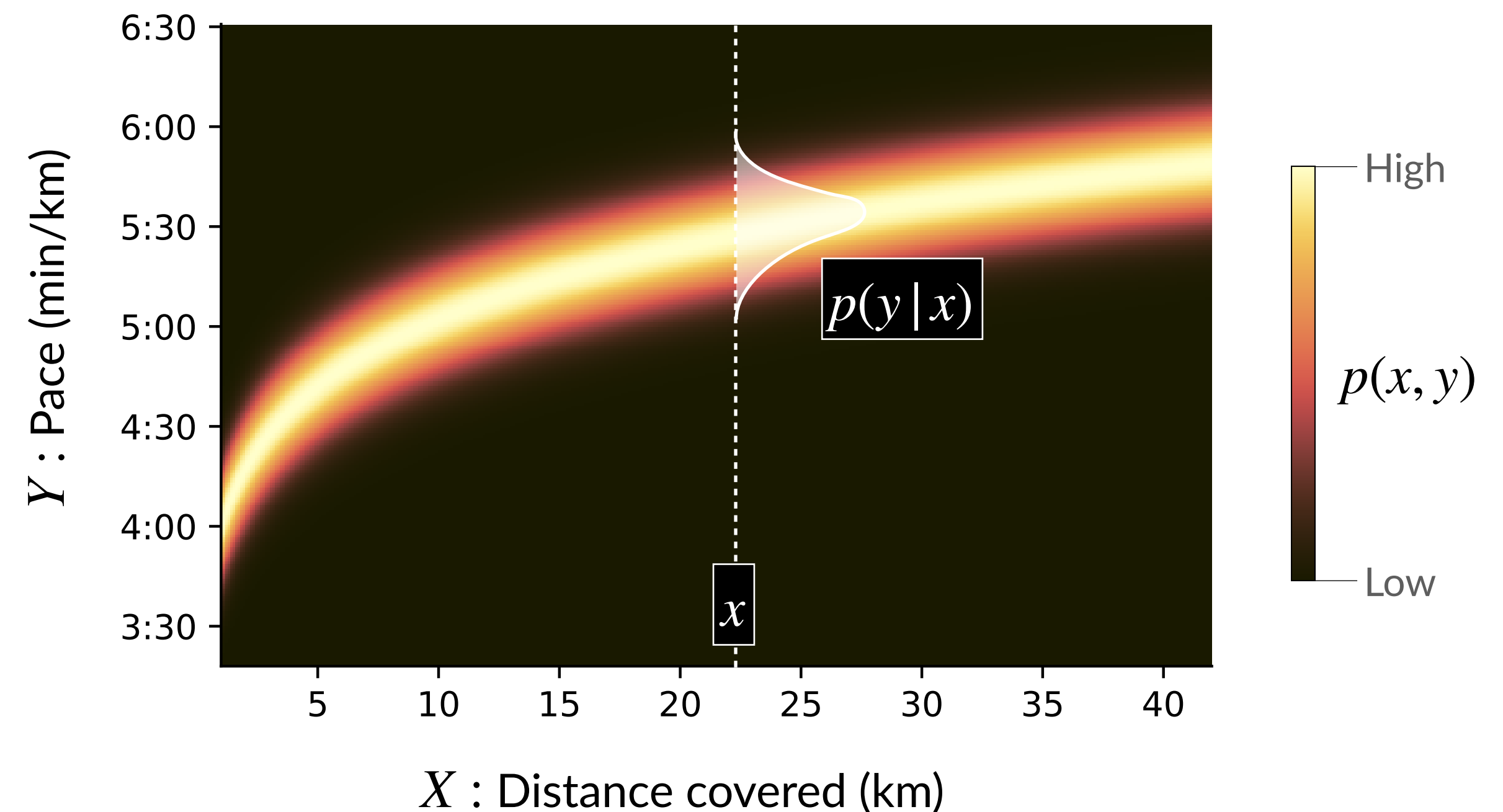
To find the minimum, we differentiate the argument:

$$\begin{aligned} \frac{d}{dc} [-2c \mathbb{E}[Y \mid X = \mathbf{x}] + c^2] &= -2 \mathbb{E}[Y \mid X = \mathbf{x}] + 2c = 0 \\ \Rightarrow c &= \mathbb{E}[Y \mid X = \mathbf{x}] \end{aligned}$$

Task

Solve the **Risk Minimisation** problem for the **Squared Loss**

Find the function f^* with the smallest expected squared miss



Risk Minimisation with Squared Loss

Regression function

Regression function

$$f^*(x) = \mathbb{E}[Y | \mathbf{X} = \mathbf{x}]$$

At each $\mathbf{x}$, the best prediction is the average of the response over the corresponding slice

Generality

No assumption has been made on the form of f or on the noise

Separability

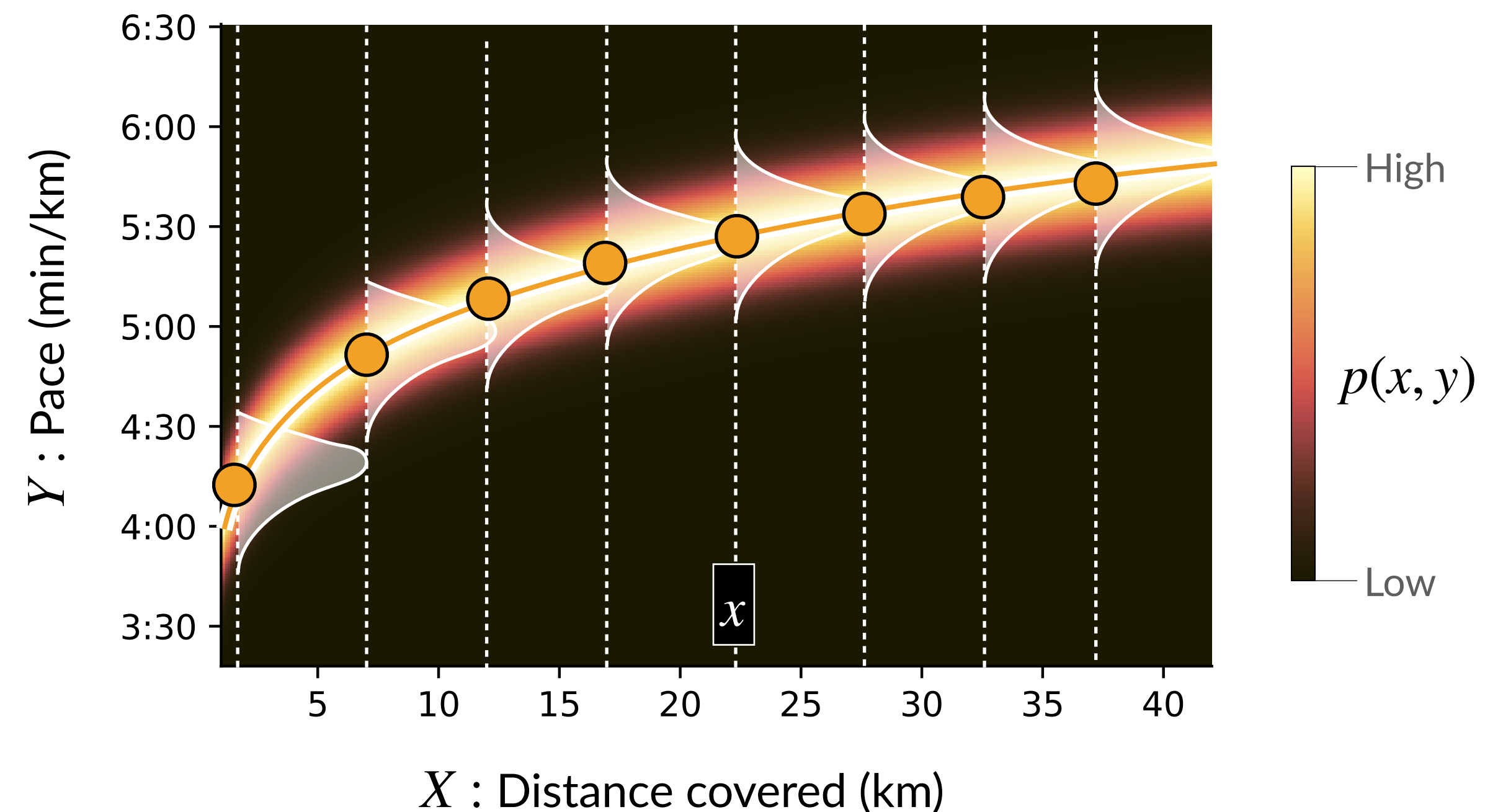
The problem splits into one independent minimisation per $\mathbf{x}$, because f is unrestricted. The choice of $f(\mathbf{x})$ at one $\mathbf{x}$ does not affect the choice at any other $\mathbf{x}$

Other losses give other targets: the absolute loss (slide 11) gives the conditional median $f^* = \text{median}(Y | \mathbf{X} = \mathbf{x})$

Task

Solve the **Risk Minimisation** problem for the **Squared Loss**

Find the function f^* with the smallest expected squared miss

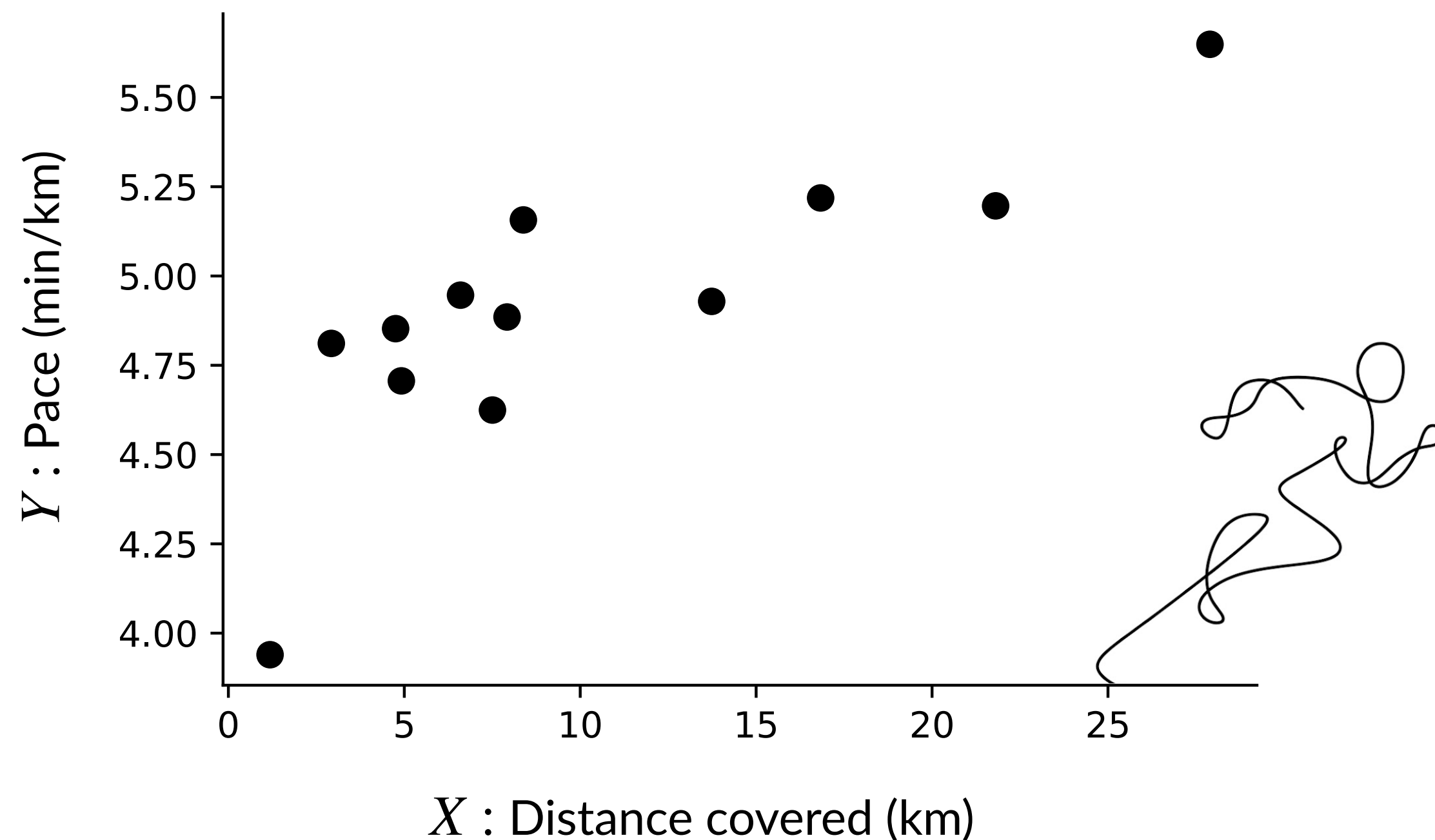


Learning from Data

The Learning Problem Setup

The discussion on the decision problem gave us a target (**regression function**) and a yardstick (**risk**). Both are defined through the joint probability, which we do not know

Let $\mathcal{D} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ be n observed pairs of the inputs and the response



* Note that $\mathcal{D}$ is the data we fit on, called the « training » data in L02. Later in the course we will revisit the train-test split

The Learning Problem

Setup

The discussion on the decision problem gave us a target (**regression function**) and a yardstick (**risk**). Both are defined through the joint probability, which we do not know

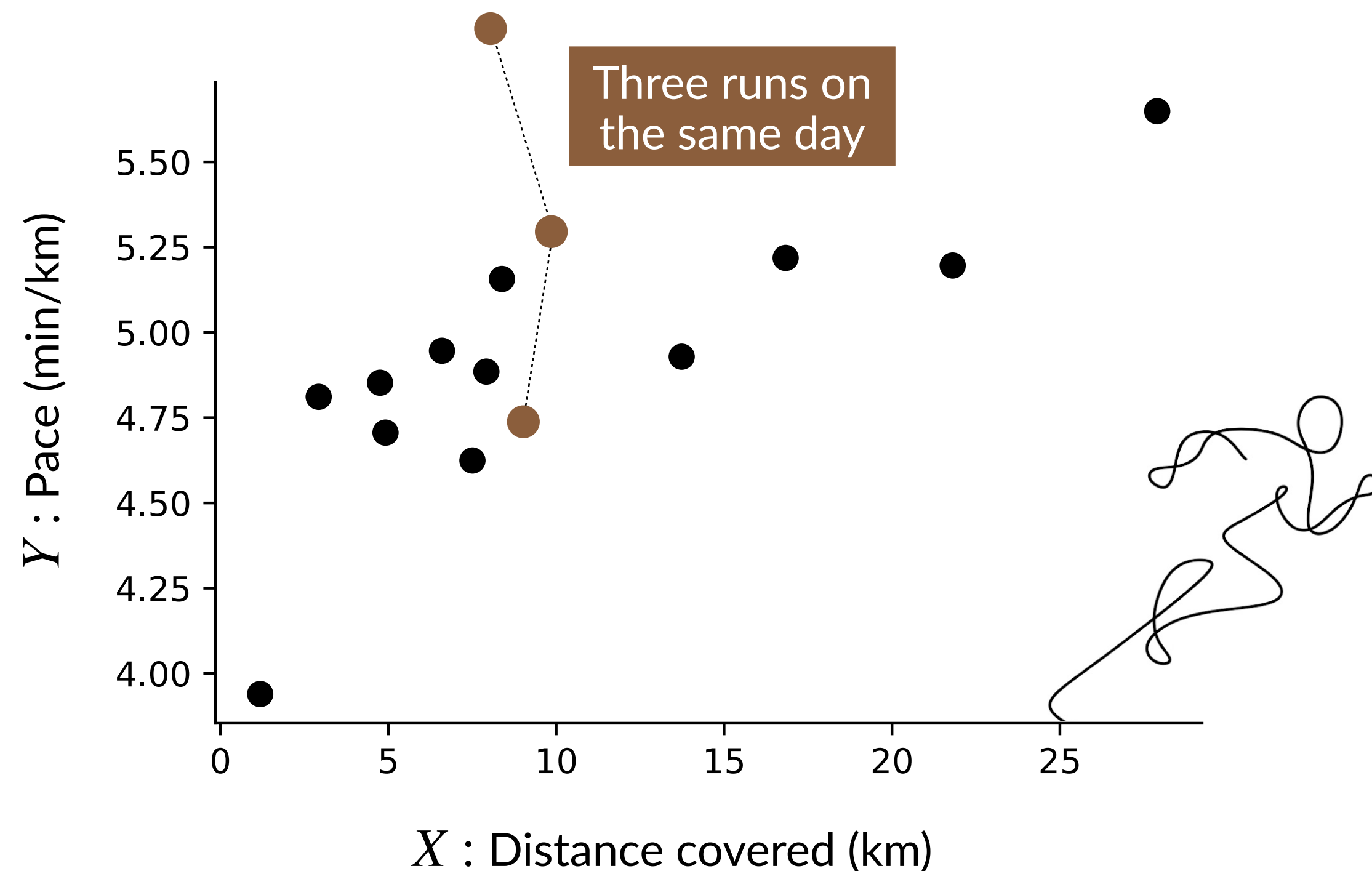
Let $\mathcal{D} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ be n observed pairs of the inputs and the response

Assumption (A1)

The pairs are **independent and identically distributed (i.i.d.)**

When it fails /

- Repeated measures
- Time series
- Unequal sampling
- Spatial effects
- ...



The Learning Problem Setup

The discussion on the decision problem gave us a target (**regression function**) and a yardstick (**risk**). Both are defined through the joint probability, which we do not know

Let $\mathcal{D} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ be n observed pairs of the inputs and the response

Task

Using $\mathcal{D}$, build a function $\hat{f}$ such that

$$R(\hat{f}) \approx R(f^*) = \min_f R(f)$$

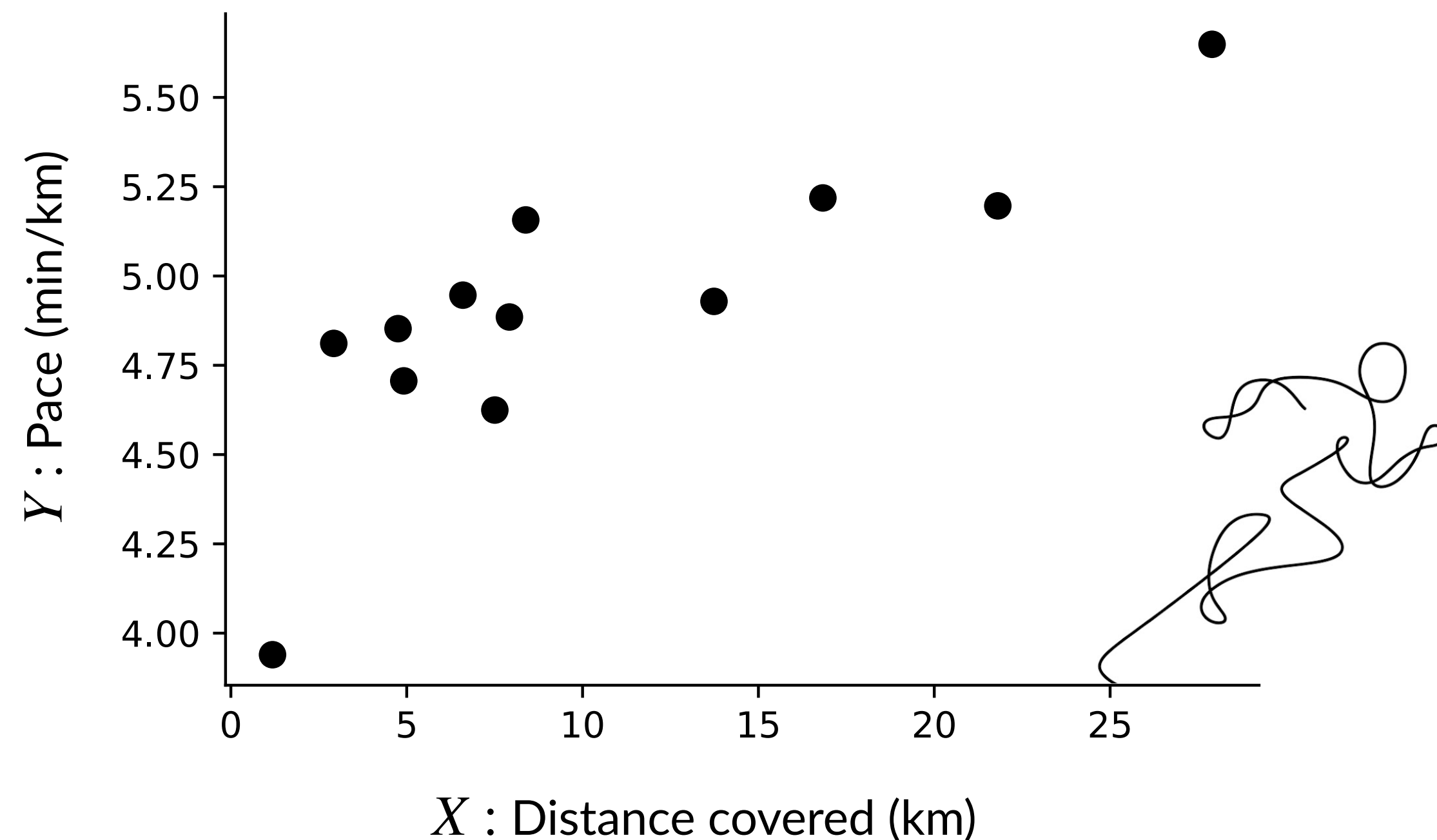
Route 1 / **Empirical Risk Minimisation (ERM)**

Replace the expectation in the risk by an average over the training data, then minimise this average

Route 2 / **Maximum Likelihood Estimation (MLE)**

Work with the distribution. Model how the response is generated, estimate the model by maximum likelihood (MLE)

Notation: a hat $\hat{}$ marks anything computed from the data



Empirical Risk Minimisation (ERM)

Direct Estimation

Given $\mathcal{D}$, a loss $\mathcal{L}$ and a family $\mathcal{F}$ of candidate functions,

$$\begin{aligned}\hat{f} &= \arg \min_{f \in \mathcal{F}} \hat{R}_n(f) \\ &= \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \mathcal{L}(y_i, f(\mathbf{x}_i))\end{aligned}$$

← « Empirical Risk » →
 $\hat{R}_n(f)$

In words / According to ERM, the best candidate in $\mathcal{F}$ is the one with the **smallest average loss on the data** $\mathcal{D}$

Empirical Risk Minimisation (ERM)

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$$= \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \mathcal{L}(y_i, f(\mathbf{x}_i))$$

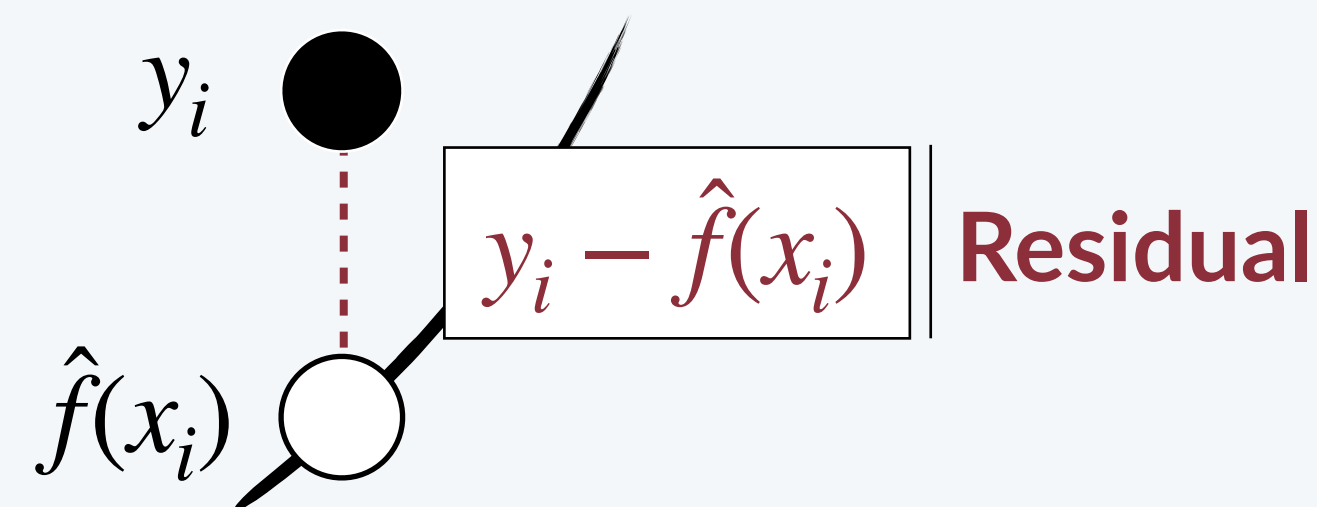
← « Empirical Risk » →
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In words / According to ERM, the best candidate in $\mathcal{F}$ is the one with the **smallest average loss on the data** $\mathcal{D}$

Example / **Squared Loss**

$$\hat{f} = \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n (y_i - f(\mathbf{x}_i))^2$$

Mean Squared
Displacement
 $\text{MSE}(f)$



Empirical Risk Minimisation (ERM)

Least Squares

Given $\mathcal{D}$, a loss $\mathcal{L}$ and a family $\mathcal{F}$ of candidate functions,

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← « Empirical Risk » →
 $\hat{R}_n(f)$

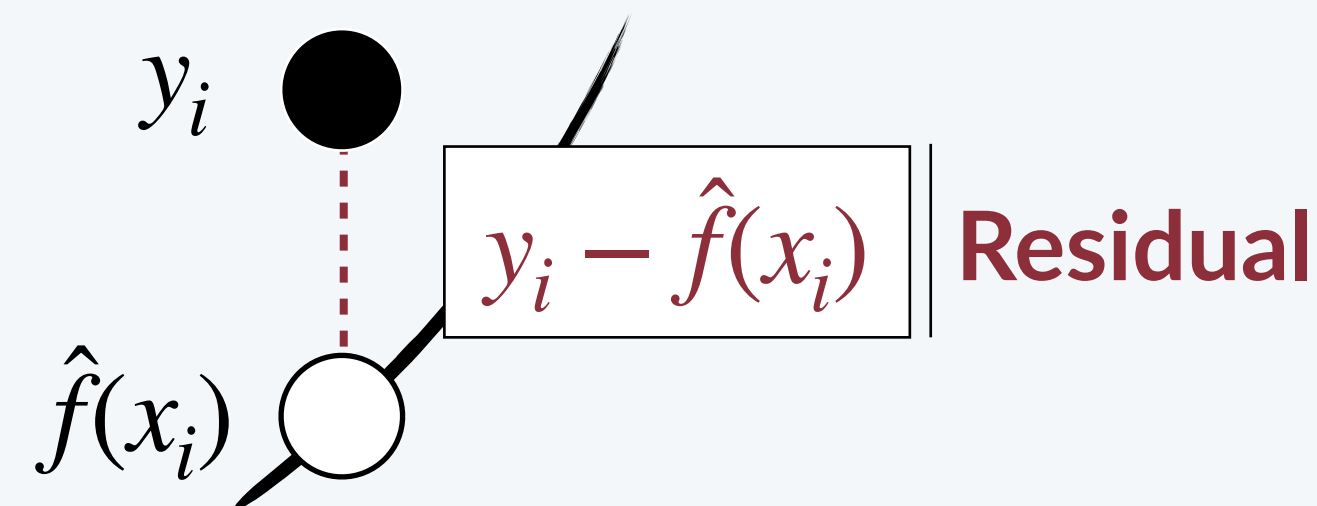
In words / According to ERM, the best candidate in $\mathcal{F}$ is the one with the **smallest average loss on the data** $\mathcal{D}$

Example / **Squared Loss**

$$\begin{aligned}\hat{f} &= \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n (y_i - f(\mathbf{x}_i))^2 \\ &= \arg \min_{f \in \mathcal{F}} \sum_{i=1}^n (y_i - f(\mathbf{x}_i))^2\end{aligned}$$

Residual Sum
of Squares
 $RSS(f)$

Least Squares
Method



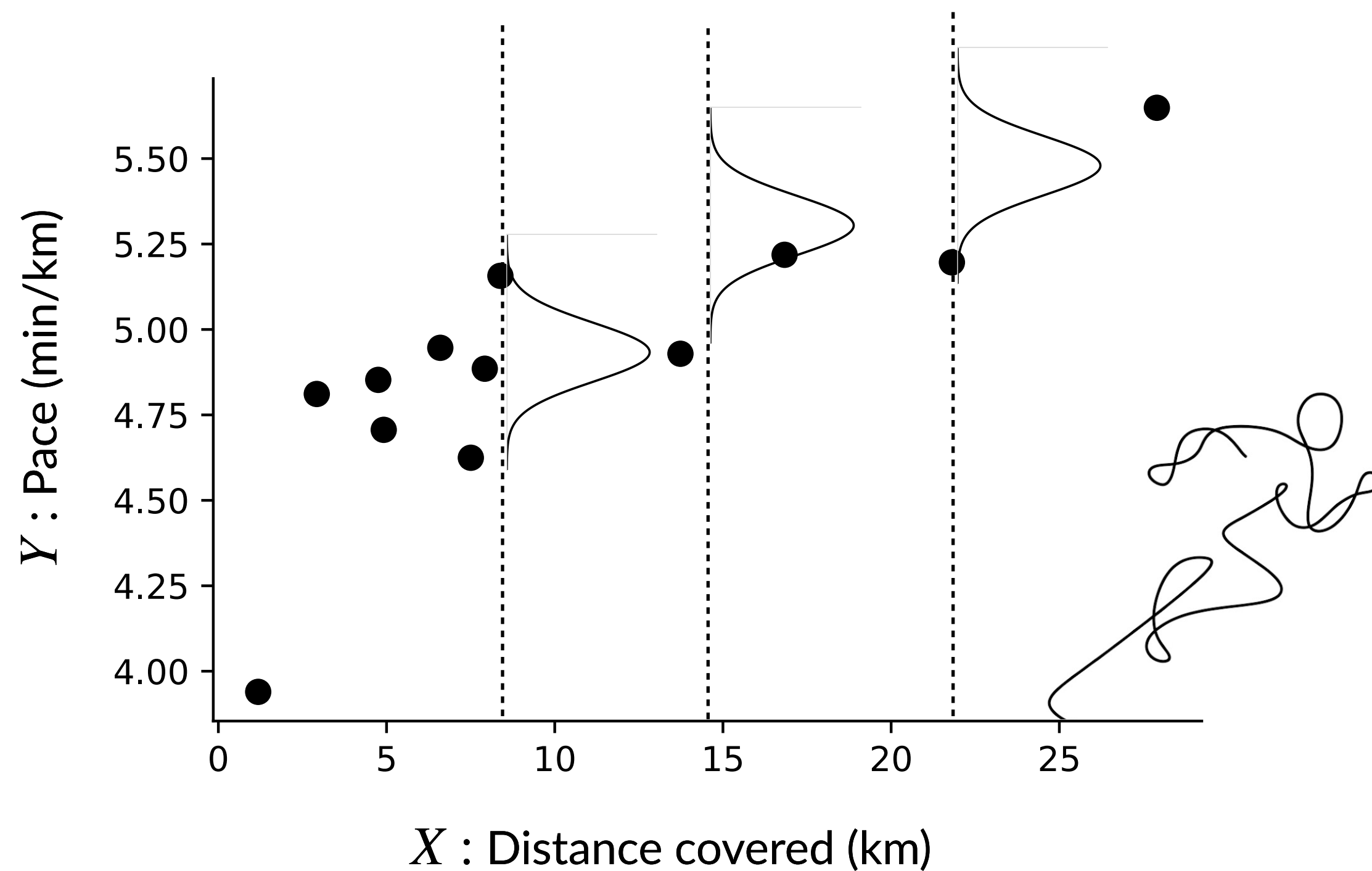
Maximum Likelihood Estimation (MLE)

Model-based Estimation



We describe how Y varies around $\mathbf{x}$, using a model with a few adjustable parameters.

We keep the parameter values under which our data look most plausible.

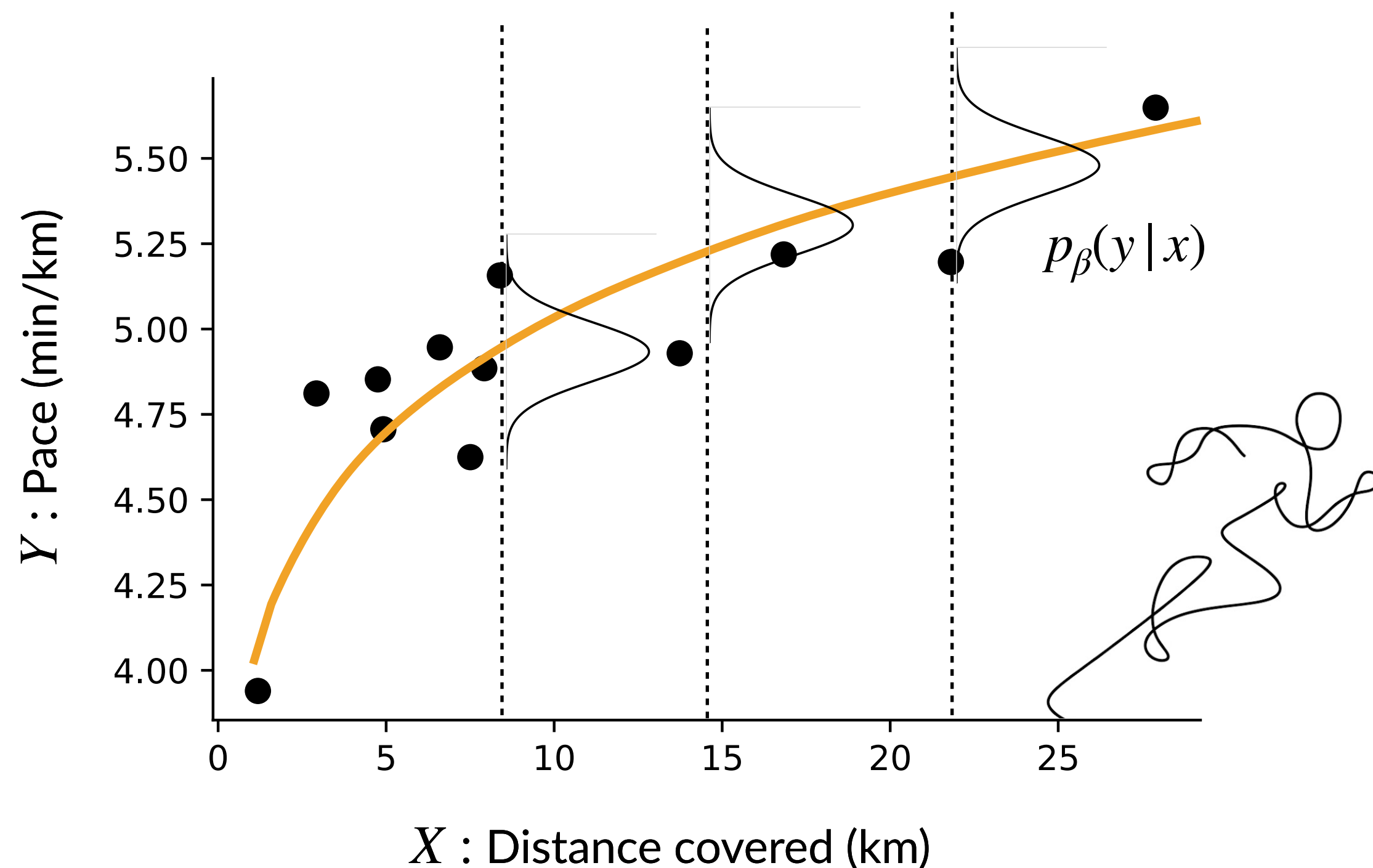


Maximum Likelihood Estimation (MLE)

Model-based Estimation



We describe how Y varies around $\mathbf{x}$, using a model with a few adjustable parameters.
We keep the parameter values under which our data look most plausible.



Restrict to **parametric models**. Given i.i.d. data $\mathcal{D}$ (A1), and a parametric family $p_{\beta}(y | x)$ of conditional densities

MLE

$$\hat{\beta} = \arg \max_{\beta \in B} L(\beta; \mathcal{D})$$

$$= \arg \max_{\beta \in B} \prod_{i=1}^n p_{\beta}(y_i | \mathbf{x}_i)$$

← Likelihood →
 $L(\beta; \mathcal{D})$

Probability of the observed responses

In words / according to maximum likelihood, the best parameter value is the one under which the **observed data are most probable**

Maximum Likelihood Estimation (MLE)

Example: Gaussian Model for Regression

$$Y = f_{\beta}(X) + \varepsilon$$

All noise distributions in the illustration have zero-mean

**Assumption
(A2)**

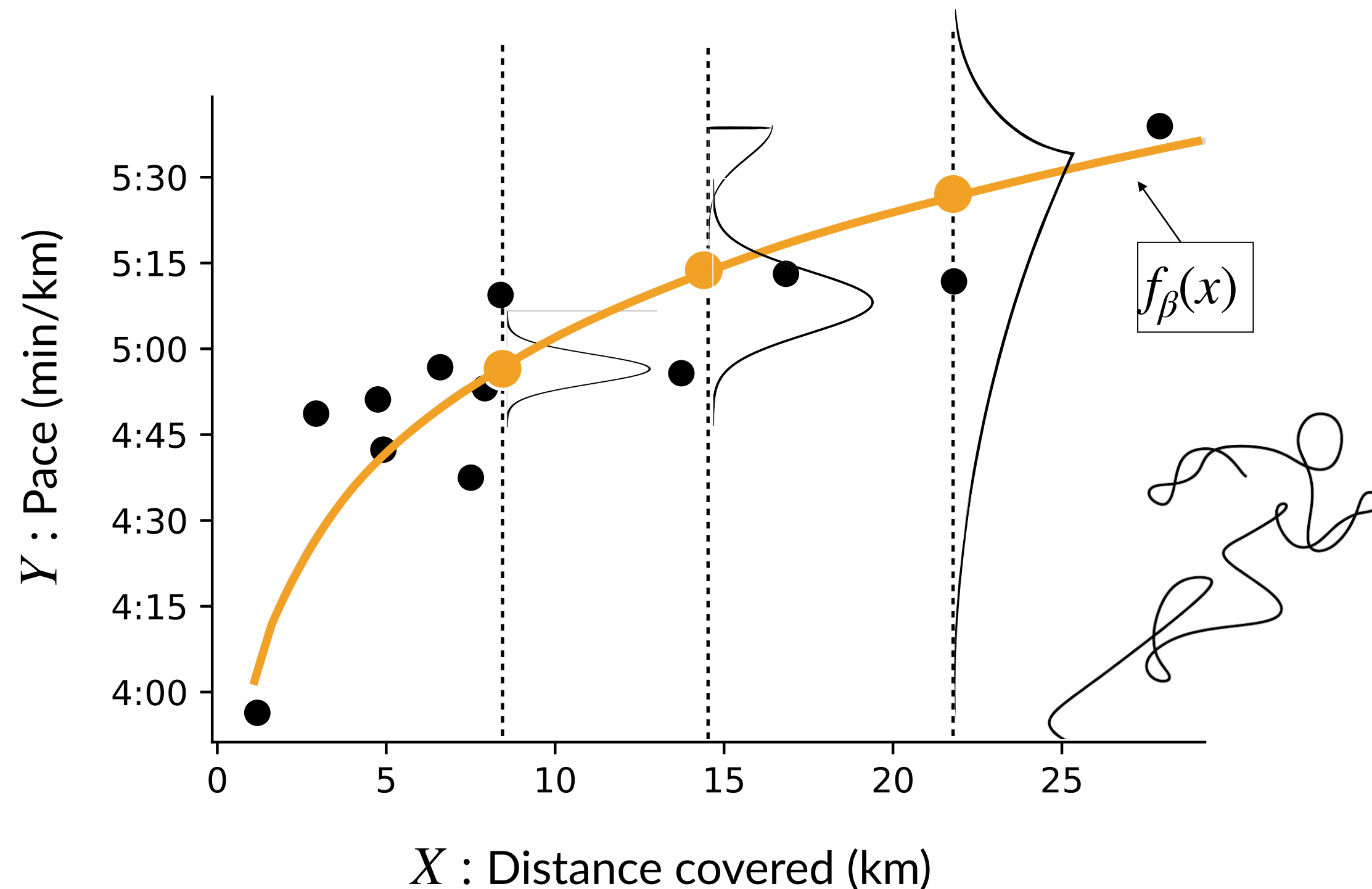
Zero-mean & additive noise

$$\mathbb{E}(\varepsilon | \mathbf{X} = \mathbf{x}) = 0$$

Note that

$$f^*(x) = \mathbb{E}(Y | X = x) = f_{\beta}(x) + \mathbb{E}(\varepsilon | X = x) = f_{\beta}(x)$$

That is, the model function is the regression function—best prediction under squared loss (Slide 19). Therefore, estimating the parameters means estimating the regression function ✓



Maximum Likelihood Estimation (MLE)

Example: Gaussian Model for Regression

$$Y = f_{\beta}(X) + \varepsilon$$

All noise distributions in the illustration have the same variance

**Assumption
(A2)**

Zero-mean & additive noise

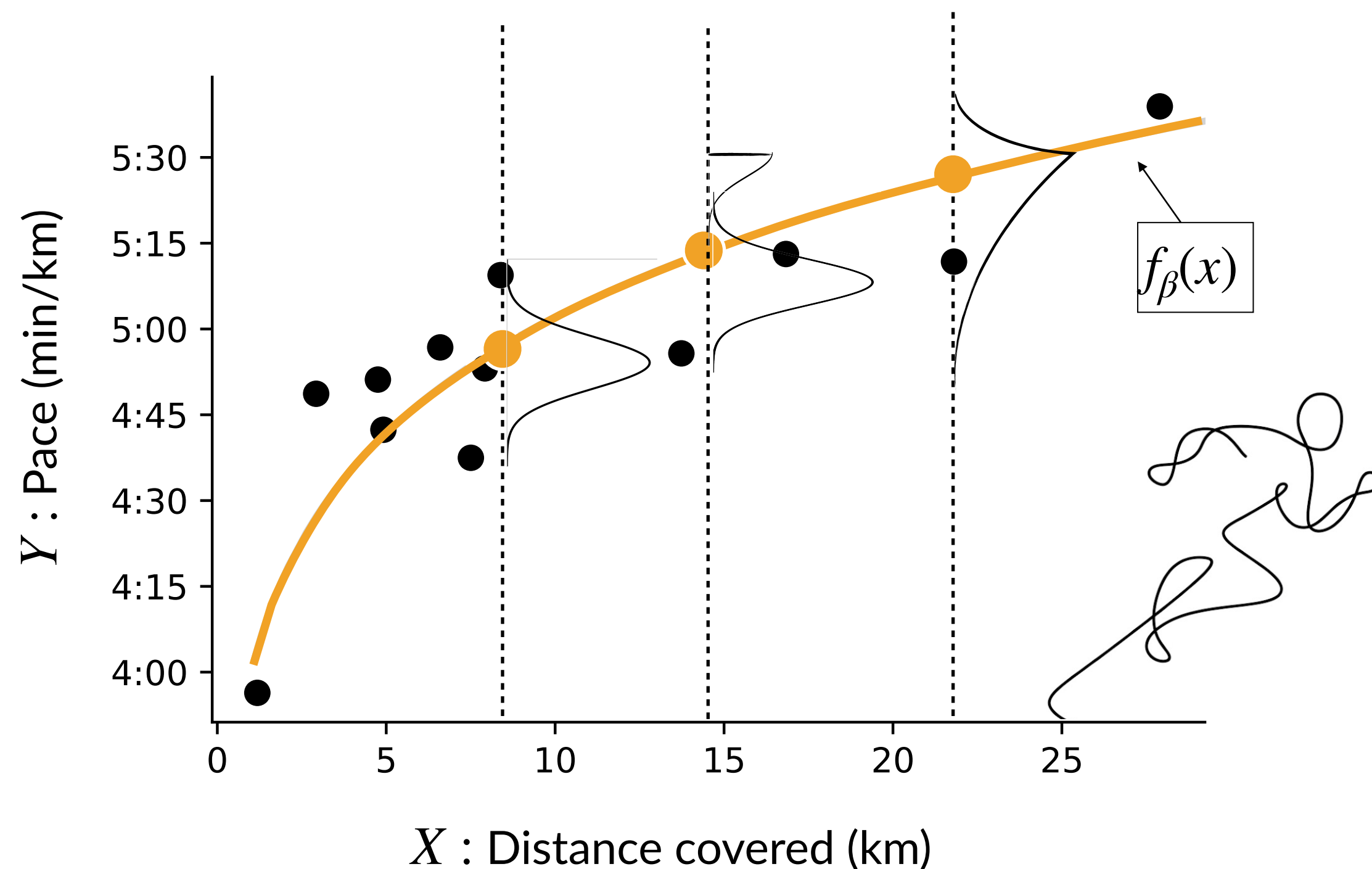
$$\mathbb{E}(\varepsilon | \mathbf{X} = \mathbf{x}) = 0$$

**Assumption
(A3)**

Constant variance

The spread of Y around its mean is the same at every $\mathbf{x}$.

$$\text{Var}(\varepsilon | X = \mathbf{x}) = \sigma^2 \quad \forall \mathbf{x}$$



Maximum Likelihood Estimation (MLE)

Example: Gaussian Model for Regression

$$Y = f_{\beta}(X) + \varepsilon$$

Assumption
(A2)

Zero-mean & additive noise

$$\mathbb{E}(\varepsilon | \mathbf{X} = \mathbf{x}) = 0$$

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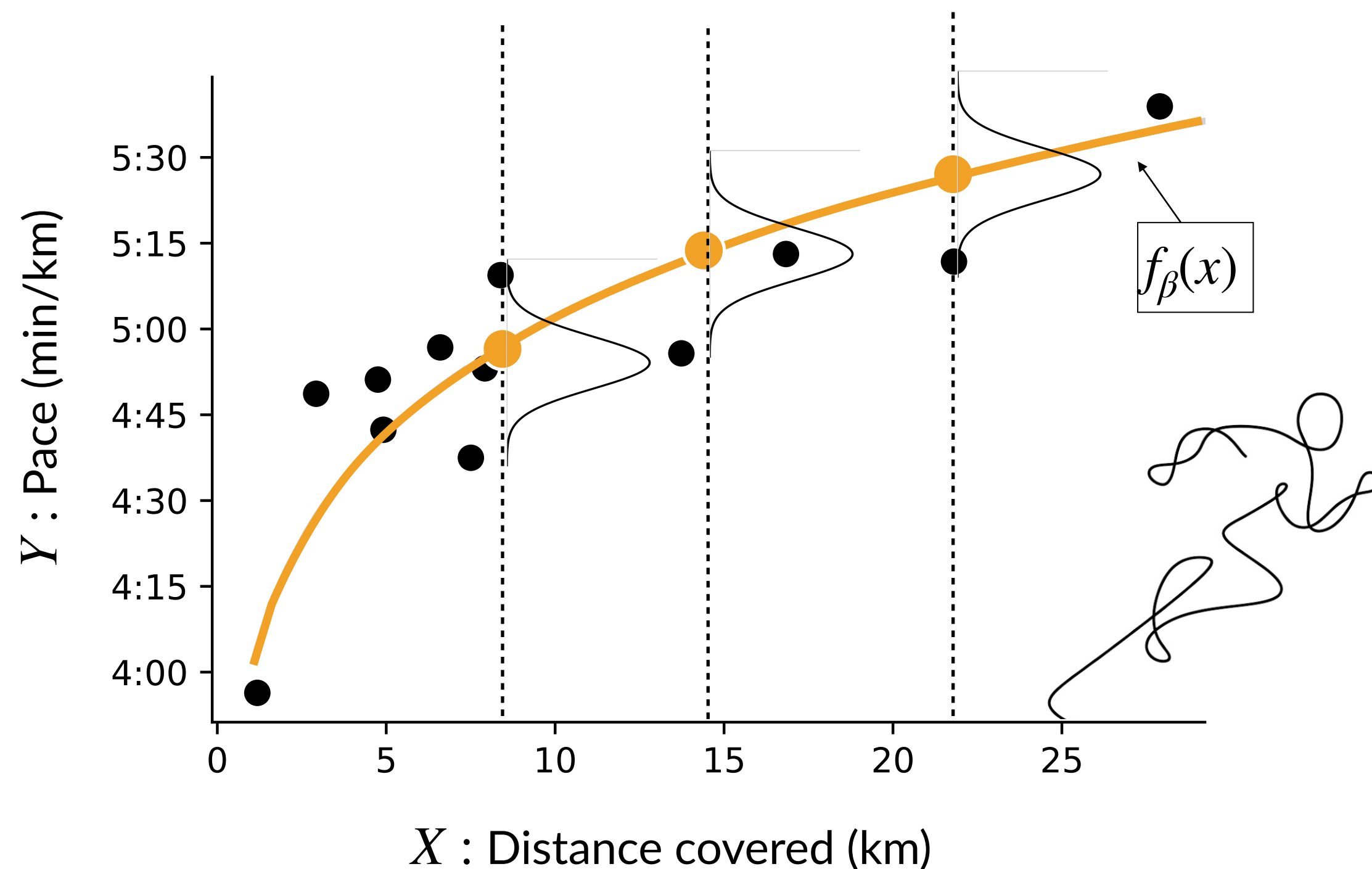
$$\text{Var}(\varepsilon | X = \mathbf{x}) = \sigma^2 \quad \forall \mathbf{x}$$

Assumption
(A4)

The noise has a **normal distribution**

The misses are symmetric around zero and large misses are very unlikely

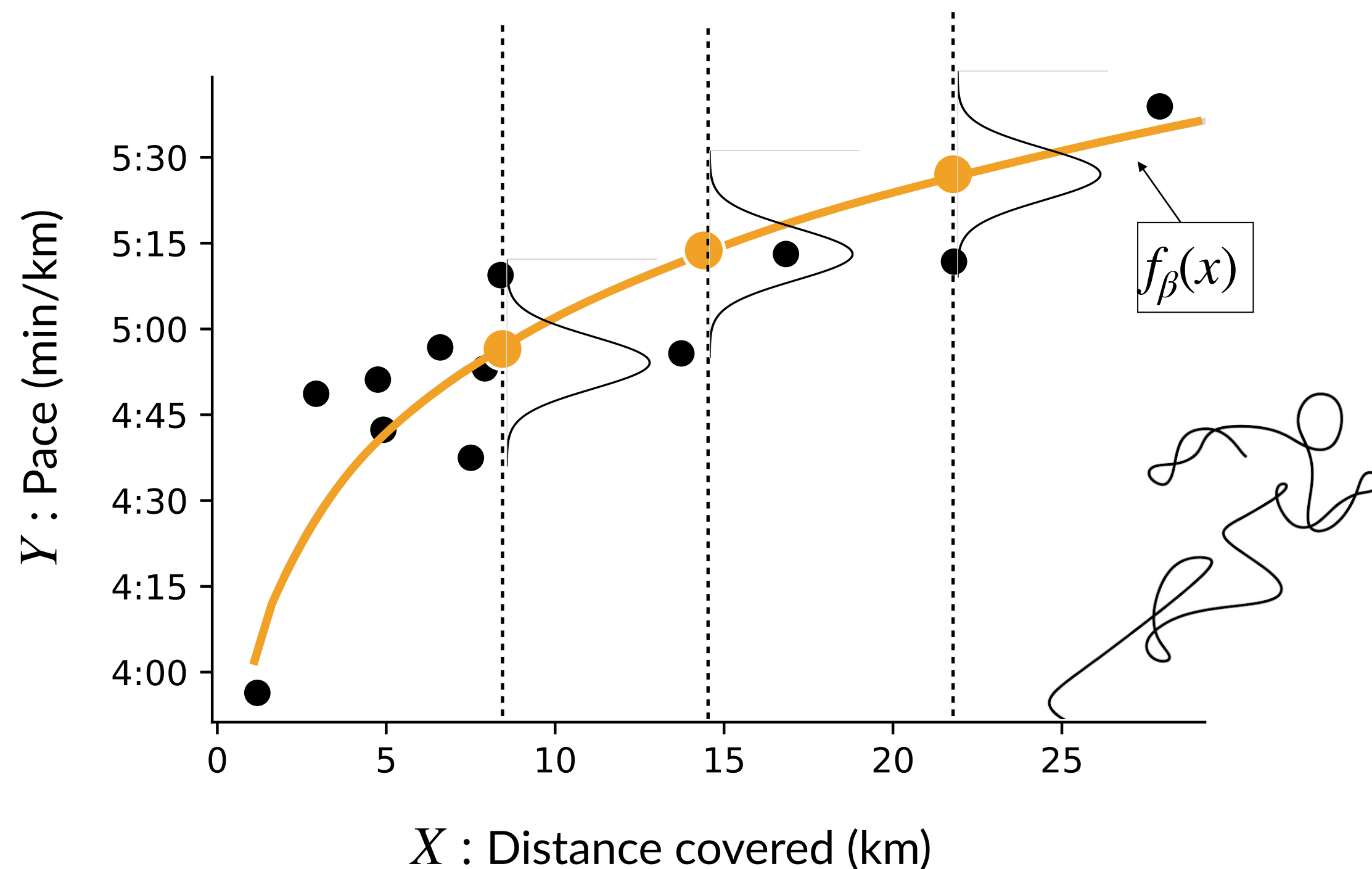
$$\varepsilon | X \sim \mathcal{N}(0, \sigma^2)$$



Maximum Likelihood Estimation (MLE)

Example: Gaussian Model for Regression

$$Y = f_{\beta}(X) + \varepsilon$$



Gaussian Model

$$p_{\beta, \sigma^2}(y \mid \mathbf{x}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y - f_{\beta}(\mathbf{x}))^2}{2\sigma^2}\right)$$

Scheme

Linear Least Squares

Lecture Report

3 : Regression



Key concepts:

- ✓ The Prediction Problem
 - Statistical Decision Theory
 - Regression Function
- ✓ The Learning Problem
 - Empirical Risk Minimisation
 - Maximum Likelihood Estimation
- ✓ Linear Least Squares
 - Analytical Solution
 -

Bibliography:

- ◎ ESL, S2.4, S2.6.3, S3.2
- ◎ DL, S4.1.2, S4.1.3, S4.1.4, S4.2
- ◎ ISL, S3.1.1, S3.1.2, S3.3

To go further:

- ◎ ESL, C2
- ◎ ISL, C3

Next week: Classification